



# Modelling Financial Markets

## Chapter 2: Market Efficiency — From EMH to Adaptive Markets



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## Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **AIC:** Akaike Information Criterion
- ▣ **AMH:** Adaptive Markets Hypothesis
- ▣ **AR:** AutoRegressive (model); in event studies: Abnormal Return
- ▣ **BDS:** Brock–Dechert–Scheinkman (test)
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BIC:** Bayesian Information Criterion
- ▣ **BIST:** Borsa İstanbul (BIST 100 index) — *the Istanbul Stock Exchange (BIST 100 index)*
- ▣ **BNR:** Banca Națională a României — *the National Bank of Romania*
- ▣ **BUX:** Budapesti Értéktőzsde index — *the Budapest Stock Exchange index*
- ▣ **BVB:** Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CAPM:** Capital Asset Pricing Model
- ▣ **CD:** Chow–Denning (multiple variance-ratio test)
- ▣ **CI:** Confidence Interval
- ▣ **CME:** Chicago Mercantile Exchange
- ▣ **CRSP:** Center for Research in Security Prices
- ▣ **DFA:** Detrended Fluctuation Analysis
- ▣ **EMH:** Efficient Market Hypothesis
- ▣ **ETF:** Exchange-Traded Fund
- ▣ **FRED:** Federal Reserve Economic Data
- ▣ **FTSE:** Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ **FWER:** Family-Wise Error Rate

## Acronyms used in this material (2/2)

- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GMM**: Generalized Method of Moments
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HK**: Hong Kong
- ▣ **IPC**: Índice de Precios y Cotizaciones — *the Mexican Stock Exchange index*
- ▣ **MDE**: Minimum Detectable Effect
- ▣ **MDH**: Martingale Difference Hypothesis
- ▣ **MIT**: Massachusetts Institute of Technology
- ▣ **ML**: Machine Learning
- ▣ **MSPE**: Mean Squared Prediction Error
- ▣ **NYSE**: New York Stock Exchange
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ **RW1**: Random Walk 1 (i.i.d. increments)
- ▣ **RW2**: Random Walk 2 (independent increments)
- ▣ **RW3**: Random Walk 3 (uncorrelated increments)
- ▣ **SD**: Standard Deviation
- ▣ **SE**: Standard Error
- ▣ **SPA**: Superior Predictive Ability (test)
- ▣ **TOM**: Turn Of the Month
- ▣ **TSMOM**: Time-Series MOMentum
- ▣ **US**: United States
- ▣ **VR**: Variance Ratio
- ▣ **WIG20**: Warszawski Indeks Giełdowy 20 — *the Warsaw Stock Exchange index of the 20 largest companies*

## Chapter Route and Outcomes

### The question of this chapter

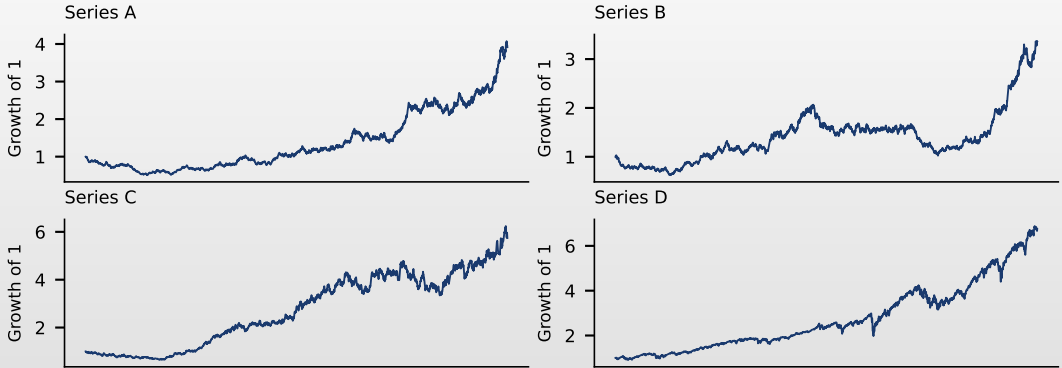
- If past prices revealed a strategy with positive expected risk-adjusted profit after costs, traders would exploit it until it disappeared
- A market in which this has already happened is called **efficient**; the claim restricts the statistical behaviour of returns, so it can be tested
- Part I: definitions, weak-form tests, predictive regressions, long memory
- Part II: adaptive markets, event studies, anomalies, behavioural finance, machine learning, a case study and one pass of the AI-assisted research loop; technical detail in the [Appendix](#)

### After this chapter you can

- state which information set a test uses and which null it actually tests (RW1, RW3, martingale difference)
- run autocorrelation, variance-ratio, long-memory and calendar tests with inference robust to volatility clustering and to multiple testing
- separate a statistical rejection from an exploitable profit and from an irrational market

## A Game: Which Series Is the Real Market? (1/2)

One of these is the S&P 500 (2010-2026); three are simulated random walks with the same drift and volatility



## A Game: Which Series Is the Real Market? (2/2)

- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?

 MFM\_ch2\_random\_walk\_game

## A Game: Which Series Is the Real Market? (2/2)

- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?
- Answer: **D**. [Roberts \(1959\)](#): random walks produce “patterns” by chance; we need statistics, not eyes
- [Bachelier \(1900\)](#) and [Samuelson \(1965\)](#): properly anticipated prices fluctuate randomly, so unpredictability is the mark of a well-functioning market

 MFM\_ch2\_random\_walk\_game

## Data Used in This Chapter

- 16 daily series up to 18 September 2026 (full list and sample starts: [▶ Appendix](#))
  - ▶ 13 equity indices from 2000: 4 developed (S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng), 9 emerging or frontier (BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai)
  - ▶ Bitcoin (from 2014; from 2011 in the yearly crypto analysis), Ethereum (from 2016) and EUR/RON, BNR reference rate (from July 2005)
- Daily log returns  $r_t = \ln P_t - \ln P_{t-1}$ , each series on its own calendar
  - ▶  $P_t$ : the closing price (index level, exchange rate) on day  $t$ ;  $\ln$ : the natural logarithm
- Equity indices: weekdays only; a close equal to the previous close is removed, since it is usually a holiday filled with the previous price
  - ▶ the rule also removes the rare genuine zero-return day; the effect on the test statistics is shown in the [▶ Appendix](#)
- Crypto-assets trade 7 days a week: 365 observations per year
- Analyses that combine several series (e.g. event studies) align prices on common trading days before computing returns

## Efficiency and the Information Set

- Fama (1970): a market is **efficient** if prices “fully reflect” all available information  $\mathcal{I}_t$ 
  - ▶ formally: no strategy using  $\mathcal{I}_t$  earns returns above the fair (risk-adjusted) return

| Form        | Information set $\mathcal{I}_t$          | Typical test (this chapter)                         |
|-------------|------------------------------------------|-----------------------------------------------------|
| Weak        | Past prices and returns                  | Autocorrelation, runs, variance ratios, long memory |
| Semi-strong | All public information (news, reports)   | Event studies                                       |
| Strong      | Public and private (insider) information | Returns of insiders and fund managers               |

- Nested: strong  $\Rightarrow$  semi-strong  $\Rightarrow$  weak; risk-adjusted returns with factor models follow in [Chapter 3](#)

## Efficiency as a Fair Game: the Martingale (1/2)

- **Fair game** for excess returns:  $\mathbb{E}[r_{t+1} - \mu_t \mid \mathcal{I}_t] = 0$ 
  - ▶  $\mathcal{I}_t$ : the information available at the end of day  $t$ ;  $\mathbb{E}[\cdot \mid \mathcal{I}_t]$ : the expectation given that information
  - ▶  $\mu_t$ : the required (equilibrium) return, the compensation for risk; in words: the return beyond  $\mu_t$  cannot be forecast from  $\mathcal{I}_t$
- **Martingale**: a process whose expected next value, given the past, equals its current value
  - ▶ With a constant required simple return  $m$ , the total-return price discounted at  $m$ ,  $P_t/(1+m)^t$ , is a martingale [Campbell, Lo & MacKinlay \(1997\)](#); discounting at the risk-free rate gives a martingale only under the risk-neutral measure
  - ▶  $P_t$  itself is one only if  $m = 0$ : a zero expected **simple** return, not a zero expected log return

## Efficiency as a Fair Game: the Martingale (2/2)

- “ $\mathbb{E}[P_{t+1} \mid \mathcal{I}_t] = P_t$ ” and “ $\mathbb{E}[\ln(P_{t+1}/P_t) \mid \mathcal{I}_t] = \mu$ ” are different restrictions
  - ▶ They differ by discounting and by the Jensen term:  $\mathbb{E}[\ln(1 + R)] \approx \mathbb{E}[R] - \text{Var}(R)/2$  for small returns ( $\mathbb{E}[R]^2$  neglected)
  - ▶  $R = P_{t+1}/P_t - 1$ : the simple return;  $\ln(1 + R)$ : the log return; the mean log return is lower than the mean simple return by about half the variance
- Consequence:  $\text{corr}(\varepsilon_{t+1}, g(\mathcal{I}_t)) = 0$  for any function  $g$  of the past,  $\varepsilon_{t+1} = r_{t+1} - \mu_t$ 
  - ▶  $\varepsilon_{t+1}$ : the unexpected part of the return; in words, no past information is correlated with tomorrow's surprise
  - ▶ In particular all autocorrelations  $\rho_k = \text{corr}(r_t, r_{t-k})$ ,  $k \geq 1$ , are zero when  $\mu_t = \mu$ ;  $k$ : the lag in days
- Efficiency does not say prices are “correct” at every moment
  - ▶ only that pricing errors are not predictable
  - ▶ it says nothing about the variance: volatility may be predictable ([Chapter 1](#))

## Random Walk Hypotheses RW1, RW2, RW3

- Log price  $p_t = \ln P_t$  with  $p_t = \mu + p_{t-1} + \varepsilon_t$ ; the three versions differ in what they assume about  $\varepsilon_t$  [Campbell, Lo & MacKinlay \(1997\)](#)
  - ▶  $\mu$ : the drift, the expected daily log return;  $\varepsilon_t$ : the shock of day  $t$ , with mean 0; so  $r_t = p_t - p_{t-1} = \mu + \varepsilon_t$
- **RW1**:  $\varepsilon_t$  i.i.d. (the strongest; rules out volatility clustering)
  - ▶ Rejected by the stylised facts of [Chapter 1](#)
- **RW2**:  $\varepsilon_t$  independent but not identically distributed (variance may change)
- **RW3**:  $\varepsilon_t$  uncorrelated:  $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ , while  $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$  is allowed
  - ▶ The relevant hypothesis for real returns: tests must be robust to heteroskedasticity

## The Joint-Hypothesis Problem

- ▣ “Fair return” requires a model of expected returns (e.g. CAPM, [Chapter 3](#))
- ▣ Every test of efficiency is a **joint** test of efficiency and of the asset-pricing model [Fama \(1991\)](#)
  - ▶ A rejection may mean an inefficient market **or** a wrong model of risk
  - ▶ Predictable **expected** returns are allowed if they compensate time-varying risk
- ▣ Some investors beat the market by luck: the question is whether success persists
- ▣ At daily horizons expected returns are tiny, so the choice of model matters little: weak-form tests are the cleanest
- ▣ Efficiency can never be “proved”, only not rejected by a given test on given data

## The Grossman–Stiglitz Paradox

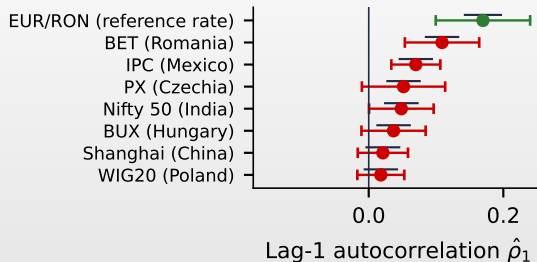
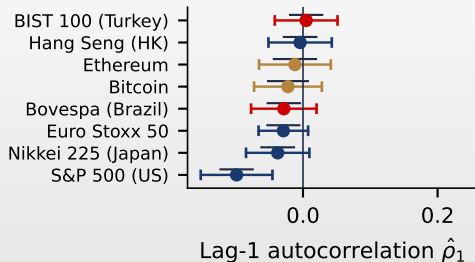
- Gathering information is costly (analysts, data, computing)
- If prices reflected **all** information, nobody would be paid for collecting it [Grossman & Stiglitz \(1980\)](#)
  - ▶ Then nobody would collect it, and prices could not reflect it
- Resolution: an **equilibrium degree of inefficiency**, just large enough to reward informed traders for their costs
- Markets with cheap information and low costs (large, liquid) should be closer to efficiency: a prediction for our comparison of developed, emerging and crypto markets
- Nor does efficiency rule out bubbles: it says they are hard to time ([Chapter 17](#))

## Autocorrelation and Its Robust Standard Error

- Sample autocorrelation:  $\hat{\rho}_k = \sum_{t=k+1}^T e_t e_{t-k} / \sum_{t=1}^T e_t^2$ , with  $e_t = r_t - \bar{r}$ 
  - ▶  $T$ : number of daily returns;  $\bar{r}$ : their mean;  $k$ : the lag;  $\hat{\rho}_k > 0$ : persistence,  $\hat{\rho}_k < 0$ : reversal
- Under RW1 (i.i.d.):  $SE(\hat{\rho}_k) \approx 1/\sqrt{T}$ ; 95% band  $\pm 1.96/\sqrt{T}$
- Unpredictable mean with volatility clustering (finite fourth moments):  $SE(\hat{\rho}_k) \approx \sqrt{\hat{\tau}_k}$ 
  - ▶  $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$  (robust variance)
  - ▶ When large moves cluster,  $\hat{\tau}_k > 1/T$ : the i.i.d. band is too narrow and classical tests reject too often; every test below uses a variance that allows for this
- S&P 500: i.i.d.  $SE = 0.012$ , robust  $SE = 0.027$ , more than twice as large

## Lag-1 Autocorrelation Across Markets (1/2)

Lag-1 autocorrelation of daily log returns across markets (full samples)



● Developed      ● Crypto      ● FX      — i.i.d. 95% band  $\pm 1.96/\sqrt{T}$       — Robust 95% interval (bars)  
 ● Emerging/frontier

## Lag-1 Autocorrelation Across Markets (2/2)

- Dots:  $\hat{\rho}_1$  with robust 95% intervals; thin segments: the naive i.i.d. band
- Positive and significant: BET 0.109 (SE 0.028), IPC 0.070, EUR/RON 0.170
- Negative and significant: S&P 500  $-0.099$
- Bitcoin  $-0.022$ , Ethereum  $-0.012$ : not significant
- Candidate explanations: thin trading (BET), a managed exchange rate (EUR/RON), short-term overreaction and liquidity provision (S&P 500)
- Economic size:  $R^2 = \hat{\rho}_1^2$ , i.e. 0.97% (S&P 500) and 1.19% (BET) of the daily variance

## Portmanteau Tests: Ljung–Box and Its Robust Version

- Ljung & Box (1978):  $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi_m^2$  under i.i.d. returns
  - ▶ tests  $H_0: \rho_1 = \dots = \rho_m = 0$  jointly;  $m$ : number of lags;  $\chi_m^2$ : chi-squared distribution with  $m$  degrees of freedom; a large  $Q$  rejects
- Robust version:  $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \rightarrow \chi_m^2$  under RW3 when the  $\hat{\rho}_k$  are asymptotically uncorrelated (condition on the slide “The Heteroskedasticity-Robust Statistic  $z^*$  below)
- $m = 10$ , 5% critical value 18.31:
  - ▶ S&P 500:  $Q = 96.3$  but  $\tilde{Q} = 19.9$  ( $p = 0.031$ )
  - ▶ BET:  $Q = 108.0$ ,  $\tilde{Q} = 24.4$  ( $p = 0.007$ )
  - ▶ Bitcoin:  $Q = 20.2$  ( $p = 0.028$ ) but  $\tilde{Q} = 11.9$  ( $p = 0.294$ )
- The robust test keeps the rejections for S&P 500 and BET, but not for Bitcoin

## The Runs Test

- Nonparametric: uses only the **signs** of returns (zero returns dropped), so it is immune to heavy tails [Wald & Wolfowitz \(1940\)](#)
- A **run** is a maximal block of equal signs: ++|-|+++ has 3 runs; Seminar A3 counts them on 20 signs
- With  $n_1$  positive and  $n_2$  negative signs,  $n = n_1 + n_2$ , under independence:
  - ▶  $\mathbb{E}[R] = \frac{2n_1n_2}{n} + 1$ ,  $\text{Var}[R] = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}$ ,  $z = \frac{R - \mathbb{E}[R]}{\sqrt{\text{Var}[R]}} \approx N(0, 1)$
  - ▶  $R$ : the observed number of runs;  $\mathbb{E}[R]$ ,  $\text{Var}[R]$ : its mean and variance if the signs were independent;  $|z| > 1.96$  rejects at 5%
- Too **few** runs ( $z < 0$ ): signs persist; too **many** ( $z > 0$ ): sign reversals; the test assumes a constant probability of a positive day
- Three markets: S&P 500  $z = 4.34$  (more runs, as  $\hat{\rho}_1 < 0$ ); BET  $z = -6.05$  (fewer runs, as  $\hat{\rho}_1 > 0$ ); Bitcoin  $z = 2.68$  although  $\hat{\rho}_1$  is not significant
  - ▶ signs and magnitudes carry different information; all markets: [Appendix](#)

## The Variance Ratio: Idea

- Uncorrelated increments with constant variance: variance grows linearly with the horizon,  
 $\text{Var}(r_t + \dots + r_{t-q+1}) = q \text{Var}(r_t)$ 
  - ▶  $q$ : the horizon in days (e.g.  $q = 5$  is one trading week)
  - ▶ With changing variances (RW2, RW3) the variance of the sum is the sum of the variances; the sample  $\widehat{\text{VR}}(q)$  still tends to 1 under the conditions of [Lo & MacKinlay \(1988\)](#)
- **Variance ratio:**  $\text{VR}(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)}$ , equal to 1 for any covariance-stationary uncorrelated series
- For a covariance-stationary series, a weighted sum of autocorrelations:
  - ▶  $\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k$
- $\text{VR}(q) > 1$ : momentum (positive autocorrelation);  $\text{VR}(q) < 1$ : mean reversion
- Worked check: S&P 500  $1 + \hat{\rho}_1 = 0.901$  vs estimated  $\text{VR}(2) = 0.901$

## The Lo–MacKinlay Test

- Lo & MacKinlay (1988): overlapping  $q$ -day sums, with bias corrections

▶  $\hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t (r_t - \hat{\mu})^2$ ,  $\hat{\sigma}_c^2(q) = \frac{1}{m} \sum_{t=q}^T \left( \sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu} \right)^2$

▶  $m = q(T - q + 1)(1 - q/T)$ ;  $\widehat{VR}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$

▶  $\hat{\mu}$ : mean daily return;  $\hat{\sigma}_a^2$ : variance of one-day returns;  $\hat{\sigma}_c^2(q)$ : variance of the overlapping  $q$ -day sums divided by  $q$ ;  $m$ : a divisor that removes the small-sample bias

- Under RW1 (homoskedastic):

▶  $z(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0, 1)$

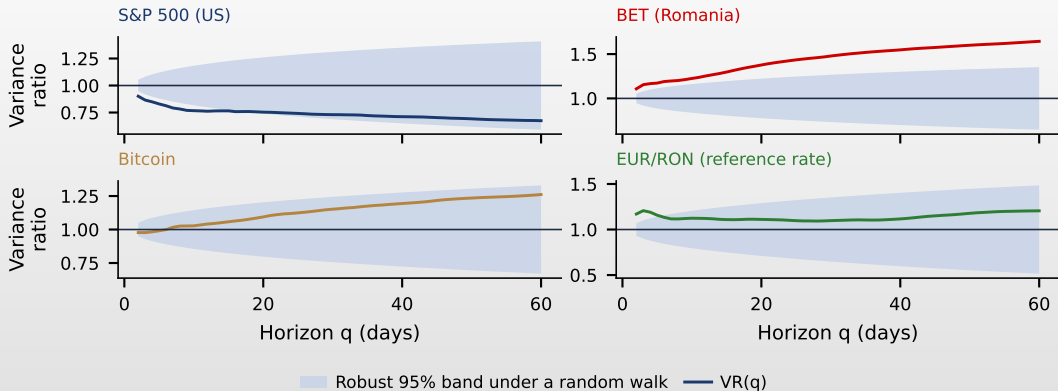
▶ the denominator is the standard error of  $\widehat{VR}(q)$  under RW1;  $|z| > 1.96$  rejects  $VR(q) = 1$  at 5%

- Homoskedastic: constant variance; this  $z$  over-rejects when volatility clusters (next slide)

## The Heteroskedasticity-Robust Statistic $z^*$

- Under RW3 plus the conditions of [Lo & MacKinlay \(1988\)](#) (finite moments, weak dependence (mixing),  $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j} \varepsilon_{t-k}] = 0$  for  $j \neq k$ ), the  $\hat{\rho}_k$  are asymptotically uncorrelated with variances  $\hat{\delta}_k / T$ 
  - ▶  $\hat{\delta}_k = T \sum_{t=k+1}^T e_t^2 e_{t-k}^2 / (\sum_{t=1}^T e_t^2)^2$  : the robust variance of  $\sqrt{T} \hat{\rho}_k$ , equal to 1 without volatility clustering
- Asymptotic variance of  $\widehat{VR}(q)$  and the test statistic:
  - ▶  $\hat{\theta}(q) = \sum_{k=1}^{q-1} \left( \frac{2(q-k)}{q} \right)^2 \hat{\delta}_k, \quad z^*(q) = \frac{\sqrt{T} (\widehat{VR}(q) - 1)}{\sqrt{\hat{\theta}(q)}} \approx N(0, 1)$
  - ▶  $2(q-k)/q$ : the weight of  $\hat{\rho}_k$  in  $\widehat{VR}(q)$ ;  $|z^*| > 1.96$  rejects  $VR(q) = 1$  at 5%
- RW3 alone is not enough: the conditions hold for GARCH with symmetric innovations, but can fail with a leverage effect
  - ▶ Without the last condition,  $\hat{\theta}(q)$  needs the full covariance matrix of the  $\hat{\rho}_k$
- S&P 500,  $q = 2$ :  $\widehat{VR} = 0.901$ ,  $z = -8.12$  but  $z^* = -3.64$ : still significant, but half as large in absolute value (Seminar B1 works it step by step)

## Variance-Ratio Profiles (1/2)



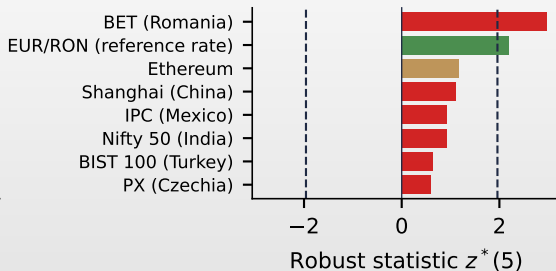
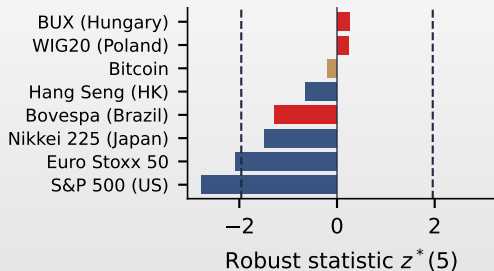
## Variance-Ratio Profiles (2/2)

- ▣  $\widehat{VR}(q)$  for  $q = 2, \dots, 60$ ; shaded: robust 95% band around 1, from  $\hat{\theta}(q)$
- ▣ S&P 500: 0.901, 0.830, 0.753, 0.674 at  $q = 2, 5, 20, 60$ : reversal, outside the band at short horizons only
- ▣ BET: 1.108, 1.173, 1.377, 1.644: persistence accumulating with  $q$
- ▣ Bitcoin: inside the band up to about one month; EUR/RON: short-run persistence of a managed float
- ▣ The band widens with  $q$ : a curve can point in one direction without being significant; direction is not evidence

 MFM\_ch2\_variance\_ratio

## Several Horizons and Markets: the Chow–Denning Test (1/2)

Variance ratio at 5 days (Lo-MacKinlay, heteroskedasticity-robust): who rejects the random walk?



■ Developed 
 ■ Emerging/frontier 
 ■ Crypto 
 ■ FX 
 --  $\pm 1.96$  (5% two-sided)

## Several Horizons and Markets: the Chow–Denning Test (2/2)

- Bars: robust  $z^*(5)$  for all 16 series; dashed lines  $\pm 1.96$
- Single horizon, reject at 5%: BET (2.98), EUR/RON (2.19), S&P 500 (−2.77), Euro Stoxx 50 (−2.08)
- Four horizons at 5% each: false-rejection probability up to  $1 - 0.95^4 = 18.5\%$ 
  - ▶ **Chow & Denning (1993)**:  $CD = \max_i |z^*(q_i)|$ , the largest of the four  $|z^*|$ , over  $q_i = 2, 5, 10, 20$ ; 5% critical value 2.49, above 1.96 to pay for the four looks
- Joint test rejects for 5 of 16: BET, S&P 500, IPC, EUR/RON, Ethereum; Bitcoin  $p = 0.835$
- No ranking “developed = efficient”: the direction differs (reversal vs persistence)

 MFM\_ch2\_variance\_ratio

## Beyond Linear Predictability: the Martingale Difference Hypothesis

- ▣ **MDH:**  $\mathbb{E}[r_t | \mathcal{I}_{t-1}] = \mu$ : no function of the past predicts the mean, linear or not
- ▣ RW3 (all  $\rho_k = 0$ ) is weaker. Counterexample:  $r_t = \varepsilon_t + \theta(\varepsilon_{t-1}^2 - \sigma^2)$ ,  $\varepsilon_t$  i.i.d. symmetric
  - ▶  $\sigma^2 = \text{Var}(\varepsilon_t)$ , so the extra term has mean 0;  $\theta \neq 0$ : how strongly yesterday's squared shock moves today's mean
  - ▶ all  $\rho_k = 0$ , yet  $\mathbb{E}[r_t | \varepsilon_{t-1}] = \theta(\varepsilon_{t-1}^2 - \sigma^2)$  is predictable: autocorrelation tests cannot see it
  - ▶ Tests of the MDH itself: the generalised spectral test [Escanciano & Velasco \(2006\)](#); an automatic portmanteau [Escanciano & Lobato \(2009\)](#) (details and results: [Appendix](#))
- ▣ **BDS** [Brock, Dechert, Scheinkman & LeBaron \(1996\)](#) tests i.i.d., not the MDH: GARCH alone makes it reject
  - ▶ A BDS rejection is evidence against RW1, not against weak-form efficiency

## Predictive Regressions and the Stambaugh Bias (1/2)

- Question: does a valuation ratio known today forecast next period's return?
- Model:  $r_{t+1} = \alpha + \beta x_t + u_{t+1}$ ,  $x_{t+1} = \theta + \rho x_t + v_{t+1}$ , with  $\text{Cov}(u, v) = \sigma_{uv}$ 
  - ▶  $r$ : excess return;  $x$ : log dividend-price ratio  $dp_t = \ln D_t - \ln P_t$ , persistent ( $\rho \approx 1$ )
  - ▶  $D_t$ : dividends paid over the last year;  $\beta$ : the predictive slope, 0 if returns are unpredictable;  $\alpha, \theta$ : intercepts
  - ▶  $\rho$ : the persistence of  $x$ ;  $u, v$ : the shocks of the two equations;  $\sigma_{uv}$ : their covariance;  $\sigma_v^2 = \text{Var}(v)$ ;  $T$ : number of periods

## Predictive Regressions and the Stambaugh Bias (2/2)

- $x_t$  is predetermined, not strictly exogenous: OLS is consistent but biased [Stambaugh \(1999\)](#)
  - ▶ Project  $u$  on  $v$ :  $u = (\sigma_{uv}/\sigma_v^2)v + e$ , so  $\hat{\beta} - \beta = (\sigma_{uv}/\sigma_v^2)(\hat{\rho} - \rho) + \text{noise}$ ;  $e$ : the part of  $u$  unrelated to  $v$
  - ▶  $\hat{\rho}$  is biased downward in small samples (Kendall), and this error passes into  $\hat{\beta}$
  - ▶ With  $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$ : 
$$\mathbb{E}[\hat{\beta} - \beta] \approx -\frac{\sigma_{uv}}{\sigma_v^2} \frac{1 + 3\rho}{T}$$
- Price up  $\Rightarrow dp$  down:  $\sigma_{uv} < 0$ , so the bias is **upward**: predictability looks stronger than it is
- Near a unit root the  $t$ -statistic is not  $N(0, 1)$  either: valid inference by a Bonferroni  $Q$ -test over a confidence interval for  $\rho$  [Campbell & Yogo \(2006\)](#)

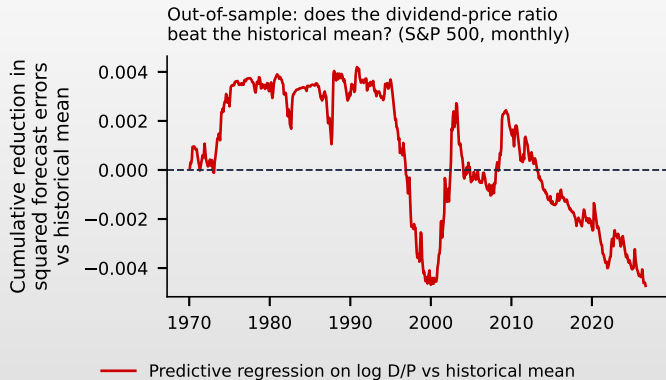
## Worked Example: the Stambaugh Bias for the S&P 500

- ▣ Monthly log excess return on the lagged  $dp_t$ , 1934–2026,  $T = 1\,111$  months (data construction: ▶ Appendix)
- ▣ OLS:  $\hat{\beta} = 0.267$  (% per month per unit of  $dp$ ), Newey–West  $t = 0.91$ : not significant even before any correction
- ▣ Inputs:  $\hat{\rho} = 0.9988$ ,  $\text{corr}(\hat{u}, \hat{v}) = -0.97$ ,  $\hat{\phi} = \hat{\sigma}_{uv} / \hat{\sigma}_v^2 = -0.925$  (as fractions, not in %)
- ▣ Substitution:  $\text{bias} = -100 \hat{\phi} (1 + 3\hat{\rho}) / T = 0.333$  (in %, like  $\hat{\beta}$ )
  - ▶ corrected slope =  $0.267 - 0.333 = -0.066$ : the whole estimate is bias
  - ▶ the correction is itself estimated: formal inference by Campbell & Yogo (2006) or a bootstrap
- ▣ Simulation with the same  $\rho$  and  $\text{corr}(u, v)$ ,  $\beta = 0$ ,  $T = 360$  months: the nominal 5%  $t$ -test rejects in 26.5% of samples

## Out-of-Sample Predictability: $R_{OS}^2$ and the Clark–West Test

- Recursive forecasts: at each month  $t$ , re-estimate on data up to  $t - 1$  only (expanding window) and forecast  $\hat{r}_t$ 
  - ▶ benchmark: the historical mean  $\bar{r}_t$  of the same data;  $R_{OS}^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t (r_t - \bar{r}_t)^2$ ;  $R_{OS}^2 > 0$ : the predictor beats the mean
  - ▶ Most predictors have  $R_{OS}^2 < 0$  [Welch & Goyal \(2008\)](#); the 2022 update confirms it [Goyal, Welch & Zafirov \(2024\)](#)
  - ▶ Pseudo out-of-sample, as in [Welch & Goyal \(2008\)](#): monthly dividends are interpolated from quarterly totals, so  $dp_t$  is not strictly real-time
- Nested models: under  $H_0$  the larger model estimates a zero  $\beta$ , which inflates its MSPE (mean squared prediction error); the Diebold–Mariano test then rejects less often than its nominal level
  - ▶ [Clark & West \(2007\)](#):  $f_t = (r_t - \bar{r}_t)^2 - [(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2]$ ; one-sided  $t$  on the mean of  $f_t$
  - ▶  $f_t$ : the squared-error gain of the predictor, plus the term  $(\bar{r}_t - \hat{r}_t)^2$  that removes the noise of estimating  $\beta$ ; a positive mean favours the predictor
- [Campbell & Thompson \(2008\)](#): impose the sign of  $\beta$  and a non-negative forecast

## Out-of-Sample Evidence: the Dividend–Price Ratio



- Cumulative squared-error gain over the historical mean, forecasts from 1970; rising line:  $dp$  helps
- 680 forecasts:  $R_{OS}^2 = -0.51\%$ ; Campbell–Thompson restrictions bind in 5.7% of months, giving  $-0.51\%$
- Clark–West  $t = 0.42$  (one-sided  $p = 0.34$ ): no significant out-of-sample predictability
- The gain peaks in 1990 and is lost in the 1990s boom, when  $dp$  fell to record lows; after 2010 it erodes again
- Economic relevance: with monthly Sharpe ratio 0.160, even  $R^2 = 0.5\%$  would raise by 20% the expected return of a mean–variance investor who times the market with the forecast (formula:

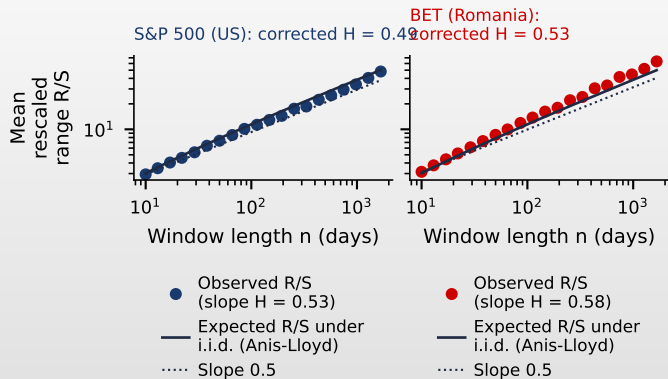
► Appendix )

 MFM\_ch2\_predictability

## The Rescaled Range (R/S) Statistic

- **Hurst (1951)** found that wet years on the Nile followed wet years more often than chance allows, using the rescaled range (history: [Appendix](#))
- For a window of  $n$  returns with mean  $\bar{r}_n$  and standard deviation  $s_n$ :
  - ▶ Cumulative deviations  $Y_k = \sum_{i=1}^k (r_i - \bar{r}_n)$ ,  $k = 1, \dots, n$
  - ▶ Range  $R_n = \max_k Y_k - \min_k Y_k$ ; rescaled range  $R_n/s_n$
- Scaling law:  $\mathbb{E}[R_n/s_n] \approx c n^H$ , where  $H$  is the **Hurst exponent**
  - ▶  $c$ : a constant; taking logs,  $\ln(R/S) \approx \ln c + H \ln n$ , so  $H$  is a slope
  - ▶ Estimate: slope of  $\ln(R/S)$  on  $\ln n$ , averaging  $R/S$  over non-overlapping windows
- $H = 0.5$ : no long memory (short-run dependence allowed);  $H > 0.5$ : persistence;  $H < 0.5$ : anti-persistence
  - ▶ **Long memory**: autocorrelations decay so slowly that their sum diverges,  $\rho_k \sim k^{2H-2}$

## R/S Analysis: S&amp;P 500 and BET



- Dots: average  $R/S$  by window length  $n$ ; line: the Anis & Lloyd (1976) expectation under i.i.d. Normal returns (exact formula: [Appendix](#))
- Small-sample bias: even for i.i.d. returns the raw slope exceeds 0.5; the Anis-Lloyd corrected exponent  $H_{AL} = 0.5 + \text{slope of } \ln(R/S) - \ln \mathbb{E}[R_n/s_n]$
- S&P 500: raw 0.530, corrected 0.486: the “persistence” was the artefact
- BET: raw 0.578, corrected 0.533: persistence remains
- Open: is BET's  $H > 0.5$  **long** memory or just  $\hat{\rho}_1 = 0.109$ ? Classical R/S cannot tell

## Hurst Exponent and Variance Ratio: One Horizon Is Not Enough

- ▣ For exactly self-similar increments (fractional Gaussian noise):  $\text{Var}(r_t + \dots + r_{t-q+1}) = q^{2H} \text{Var}(r_t)$
- ▣ Hence  $\text{VR}(q) = q^{2H-1}$  and the **implied**  $H(q) = 0.5 + \ln \text{VR}(q) / (2 \ln q)$ : BET 0.553, S&P 500 0.453 at  $q = 20$
- ▣ Short-memory counterexample: an AR(1) with  $\rho_1 = -0.1$ ,  $\rho_k = (-0.1)^k$ 
  - ▶  $\text{VR}(20) = 0.826$ , so  $H(20) = 0.468$ : “anti-persistent” without any long memory
  - ▶ yet  $H(250) = 0.482$ : for short memory  $H(q) \rightarrow 0.5$  as  $q \rightarrow \infty$  (Seminar A5–A6)
  - ▶ an AR(1) with BET’s  $\hat{\rho}_1 = 0.109$  gives  $H(20) = 0.535$ : most of BET’s value is short memory
- ▣ Separating the two needs a statistic built for it: Lo’s modified R/S (next slide)

## Lo's Modified R/S Test

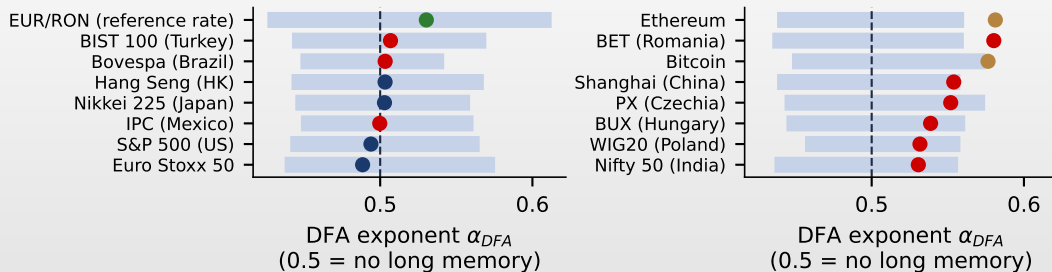
- ▣ Lo (1991): short-run autocorrelation inflates R/S and mimics long memory
- ▣ Replace  $s_n$  by a long-run (HAC) standard deviation with  $\ell$  lags:
  - ▶  $\hat{\sigma}_\ell^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^{\ell} \left(1 - \frac{j}{\ell+1}\right) \hat{\gamma}_j$ ,  $\hat{\gamma}_j$ : autocovariance at lag  $j$  ( $\hat{\gamma}_0$ : the variance), with declining weights
  - ▶  $V = \frac{\max_k Y_k - \min_k Y_k}{\hat{\sigma}_\ell \sqrt{T}}$ ;  $Y_k$ : cumulative deviations over the full sample of  $T$  returns; bandwidth  $\ell$  chosen as in Andrews (1991)
- ▣ Under short memory,  $V$  lies in  $[0.809, 1.862]$  with probability 95%
- ▣ Full samples: S&P 500  $V = 1.559$ , Bitcoin 1.547, PX 1.750, EUR/RON 1.429: not rejected
  - ▶ BET  $V = 1.961$  ( $\ell = 7$ ) and Ethereum  $V = 1.980$ : just above 1.862, weak evidence of long memory

## Detrended Fluctuation Analysis (DFA)

- Peng et al. (1994): linear detrending of the profile removes a locally constant mean of returns; trends in returns need higher-order DFA
- Algorithm (Seminar B3 shows one window step by step):
  - ▶ Profile  $y_k = \sum_{i=1}^k (r_i - \bar{r})$ ; split into boxes of length  $n$
  - ▶ In each box fit a linear trend and keep the residuals
  - ▶  $F(n)$  = root mean square of all residuals; repeat for many  $n$
- $F(n) \propto n^{\alpha_{DFA}}$ : the DFA exponent  $\alpha_{DFA}$  plays the role of  $H$  ( $\alpha_{DFA} = 0.5$ : no long memory)
- Null band by **wild bootstrap**:  $e_t \eta_t$ ,  $\eta_t = \pm 1$  with probability 1/2 (199 replications);  $e_t = r_t - \bar{r}$ ,  $\eta_t$ : a random sign
  - ▶ Keeps the volatility path  $|e_t|$  (allowed under RW3) and destroys serial correlation; shuffling would destroy both
  - ▶ Implicit null: signs symmetric and independent given  $|e_t|$ , stronger than RW3: a benchmark, not an exact RW3 test and not a confidence interval for  $\alpha_{DFA}$

## DFA Exponent Across Markets (1/2)

Long memory in returns? Detrended fluctuation analysis of daily log returns

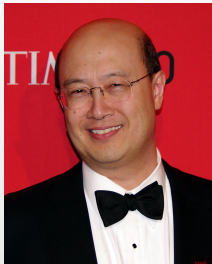


● Developed 
 ● Emerging/frontier 
 ● Crypto 
 ● FX 
  Wild-bootstrap 95% band (RW3 null)

## DFA Exponent Across Markets (2/2)

- Dots:  $\alpha_{DFA}$ ; bars: sign-flip 95% null band (volatility kept, no autocorrelation)
- Inside the band: S&P 500 0.494, Euro Stoxx 50 0.488, Nikkei 0.503, PX 0.552, Shanghai 0.554
- Above (3 of 16): Ethereum 0.581, BET 0.580 (band to 0.561), Bitcoin 0.576, borderline (band to 0.575)
- A shuffled band flags 5: PX and Shanghai were “memory” only through volatility clustering
- Cross-checks: Lo’s  $V$  rejects BET and Ethereum marginally, not Bitcoin; the local Whittle estimator of the long-memory parameter  $d = H - 0.5$  for BET:  $\hat{d} = 0.118$ , 95% CI [0.062, 0.174] ([Appendix](#))
- Lesson: memory in returns is weak and fragile; memory in volatility ([Chapter 1](#)) is strong

## The Adaptive Market Hypothesis



Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0;

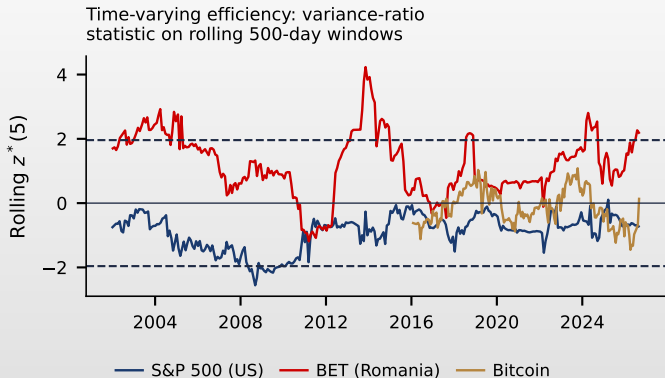
Wikimedia Commons

- Lo (2004); book-length version Lo (2017)
- Markets are an ecology of competing species (traders, funds, algorithms) that adapt by trial and error, not by perfect optimisation
  - ▶ efficiency is a degree that depends on the number of competitors, the profit opportunities and the environment
- Prediction: efficiency **varies over time and with market conditions**; anomalies appear, weaken and may reappear
- Caution: time-varying statistics alone do not identify the AMH, since an EMH with time-varying risk premia also allows them (comparison table:  
▶ [Appendix](#) )
- Evidence of time variation: survey [Lim & Brooks \(2011\)](#); long-run indices [Urquhart & McGroarty \(2016\)](#)

## Measuring Time-Varying Efficiency

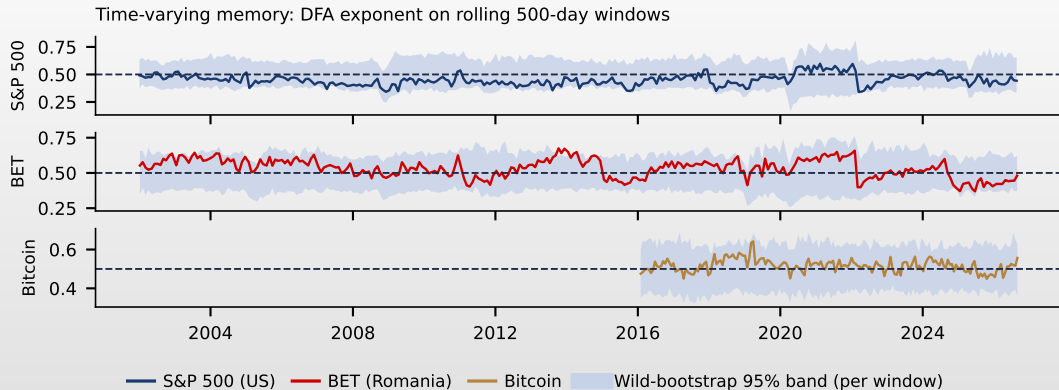
- Compute a statistic on a **rolling window**: 500 trading days (about two years), moved forward by 21 days (one month)
  - ▶ Statistics:  $z^*(5)$  of the variance-ratio test and the DFA exponent
- Null band for DFA: wild bootstrap **inside each window** (199 sign flips), so the band follows the window's volatility
  - ▶ An i.i.d. Student- $t_4$  band  $[0.378, 0.614]$  ignores volatility clustering, which RW3 allows and which raises  $\alpha_{DFA}$
- Reading the results
  - ▶ Overlapping windows are strongly correlated: runs of rejections are not independent evidence
  - ▶ Even under the null about 5% of windows reject; a share near 5% is a description, not a calibrated test of stability

## Rolling Variance Ratio: S&P 500, BET, Bitcoin



- $z^*(5)$  on rolling 500-day windows; dashed lines  $\pm 1.96$
- Share of windows with  $|z^*(5)| > 1.96$ : S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- S&P 500: the full-sample rejection sits in a few turbulent windows
- BET: rejections cluster in episodes, then fade; FTSE upgrade to Secondary Emerging effective 21 Sep 2020, a candidate change in the investor base (Seminar C1)
- Bitcoin, 2014–2026: no window rejects

## Rolling DFA Exponent: S&amp;P 500, BET, Bitcoin (1/2)



## Rolling DFA Exponent: S&P 500, BET, Bitcoin (2/2)

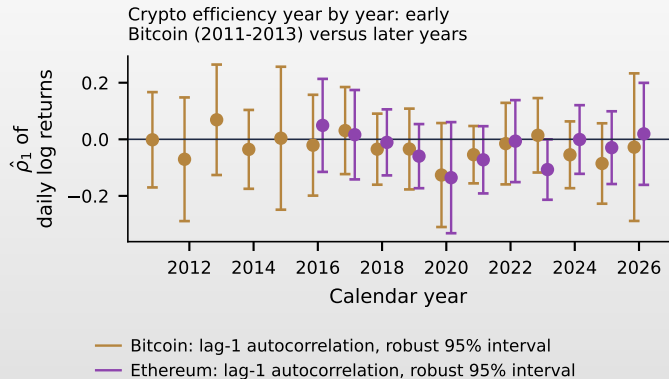
- $\alpha_{DFA}$  on rolling 500-day windows; shaded: wild-bootstrap 95% band computed in each window
- Outside their own band: S&P 500 2.0%, BET 2.7%, Bitcoin 0.0% of windows
- With the fixed i.i.d. Student- $t_4$  band: 6.4%, 12.9%, 1.1%: the apparent BET “episodes” were volatility clustering
- Low power: the BET band averages [0.36, 0.64]; 500 observations cannot pin down a scaling exponent

 MFM\_ch2\_rolling\_efficiency

## The AMH, Tested

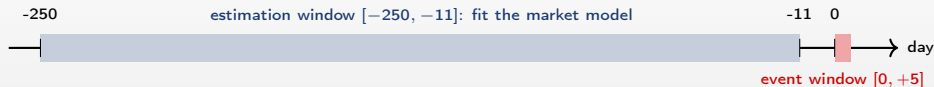
- Kim, Shamsuddin & Lim (2011): century-long US data; predictability varies with market conditions (crises, bubbles), not along a trend
  - ▶ This turns the AMH narrative into a regression with testable coefficients
- Our version: regress the rolling  $z^*(5)$  on the window's annualised volatility (%) and a crisis dummy (window overlaps Sep 2008–Jun 2009 or Feb–Jun 2020)
  - ▶ Newey–West errors with 24 lags: consecutive windows share 95% of their days
- S&P 500: crisis  $-0.67$  ( $t = -2.37$ ), volatility  $0.006$  ( $t = 0.27$ ),  $R^2 = 0.20$
- BET: crisis  $-0.82$  ( $t = -2.77$ ), volatility  $-0.008$  ( $t = -0.36$ ),  $R^2 = 0.15$
- Crisis windows are associated with lower  $z^*(5)$  (towards reversal) in both markets, beyond volatility: a conditional association, not a causal effect
  - ▶  $z^*$  mixes  $\widehat{\sqrt{R}} - 1$  and its standard error; state dependence is consistent with the AMH but does not identify it uniquely

## Crypto Efficiency Year by Year



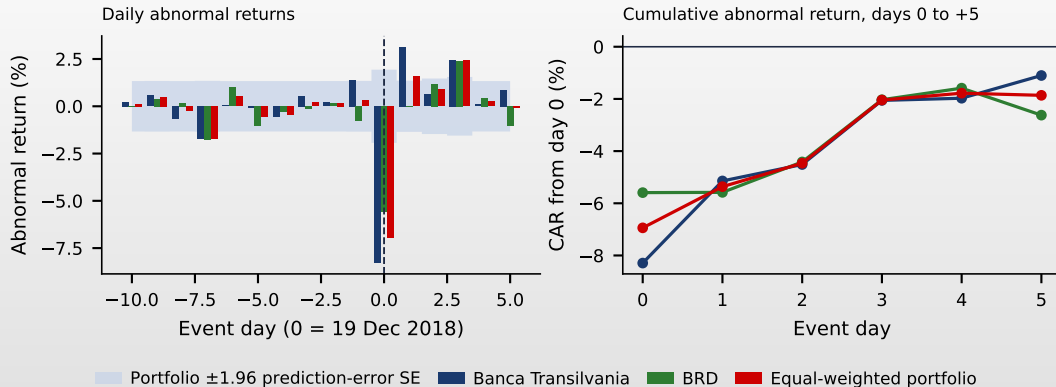
- Lag-1 autocorrelation for each calendar year, robust 95% intervals; Bitcoin from 2011
- Only 1 of 27 yearly  $\hat{\rho}_1$  has  $|t| > 1.96$  (Ethereum 2023,  $-0.107$ ), against  $0.05 \times 27 = 1.35$  expected by chance
- Unlike [Urquhart \(2016\)](#), who rejects efficiency in the early years: other tests (including BDS), another price source and classical errors (literature: [Appendix](#))
- Power: one year of 365 returns gives a robust SE of about 0.06–0.13, so small effects are invisible

## The Event-Study Method



- Fama, Fisher, Jensen & Roll (1969): how fast do prices absorb public news (stock splits)?
- Market model:  $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$ ,  $r_{m,t}$  the market return; abnormal return  $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$ ;  
 $CAR_i[\tau_1, \tau_2] = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$
- First-pass test:  $t = CAR_i / (\hat{\sigma}_\varepsilon \sqrt{L})$ ,  $L$  the event-window length MacKinlay (1997)
  - ▶  $a_i, b_i$ : intercept and market beta of firm  $i$ , fitted in the estimation window;  $\hat{\sigma}_\varepsilon$ : the residual standard deviation there;  $[\tau_1, \tau_2]$ : the days summed
  - ▶  $AR$ : the return beyond what the market move predicts;  $\sqrt{L}$ : the SD of a sum of  $L$  independent residuals grows with  $\sqrt{L}$
- For unanticipated news, efficiency predicts a jump on day 0 and no predictable drift or reversal afterwards
- Daily data can only show absorption within a day; speed measured in microseconds needs intraday data (Chapter 10; literature: [▶ Appendix](#))

## Worked Example: the Romanian Bank Tax of December 2018 (1/2)



## Worked Example: the Romanian Bank Tax of December 2018 (2/2)

- Tax announced on the evening of 18 Dec 2018, day 0 = 19 Dec; Banca Transilvania (TLV), BRD and their equal-weighted portfolio vs BET (Seminar 0, B7)
- TLV:  $\hat{b} = 1.18$ ,  $\hat{\sigma}_\varepsilon = 0.83\%$ ,  $AR_0 = -8.3\%$  ( $t = -10.0$ );  $CAR[0, +5] = -1.1\%$  ( $t = -0.54$ ), since  $CAR[+1, +5] = +7.2\%$  ( $t = 3.82$ )
- Most of the day-0 fall reverses within the week: an **overreaction**, against semi-strong efficiency, or later news (the tax was softened); one event cannot separate them

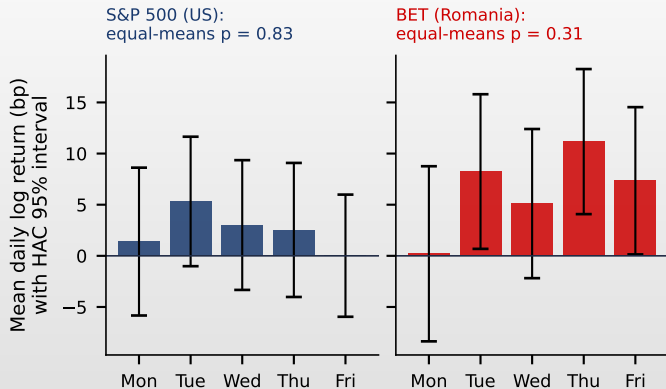
## Event Studies: Inference Done Properly

- Prediction-error variance:  $\text{Var}(AR_\tau) = \sigma_\varepsilon^2 \left[ 1 + \frac{1}{T_0} + \frac{(r_{m,\tau} - \hat{\mu}_m)^2}{(T_0 - 1)\hat{\sigma}_m^2} \right]$ ,  $T_0$  = estimation days [MacKinlay \(1997\)](#)
  - ▶  $\hat{\mu}_m, \hat{\sigma}_m^2$ : mean and variance of the market return in the estimation window; the bracket grows when the event-day market move is unusual
  - ▶ Bank tax, BET  $r_{m,0} = -11.9\%$ : the bracket is 2.12, so TLV's  $t$  shrinks in absolute value from  $-10.0$  to  $-6.85$ ; BRD:  $AR_0 = -5.6\%$ ,  $t = -4.03$
- Several firms: standardise each  $AR$  by its own prediction error and use the cross-sectional variance, robust to event-induced variance [Boehmer, Masumeci & Poulsen \(1991\)](#)
- Same event day for all firms: abnormal returns are cross-correlated and cross-sectional tests over-reject; the equal-weighted portfolio test already contains the correlation
  - ▶ TLV and BRD:  $\bar{r} = 0.15$ ; mean  $AR_0$  with an independence  $t = -7.55$ , portfolio  $t = -7.03$
  - ▶  $\bar{r}$ : the average correlation between the firms' abnormal returns;  $N$ : number of firms
  - ▶ [Kolari & Pynnönen \(2010\)](#) correct a cross-sectional test by  $\sqrt{(1 - \bar{r}) / (1 + (N - 1)\bar{r})}$ ; with ten such firms the test would understate the variance of the mean  $AR$  2.81 times

## Anomalies and Calendar Effects

- An **anomaly** is a return pattern the asset-pricing model cannot explain: mispriced risk, behavioural bias or data mining
  - ▶ Families: calendar, time-series (index momentum), cross-sectional (size, value, momentum: [Chapter 3](#))
  - ▶ Returns 26% lower out of sample and 58% lower after publication [McLean & Pontiff \(2016\)](#)
- Calendar effects: Monday [French \(1980\)](#); January [Rozeff & Kinney \(1976\)](#); turn of the month (TOM: last trading day and first three of the next month) [Ariel \(1987\)](#); [Lakonishok & Smidt \(1988\)](#)
  - ▶ Day of week:  $r_t = \sum_{d=1}^5 \beta_d D_{d,t} + u_t$ , Wald test of  $\beta_1 = \dots = \beta_5$ ; TOM and January: one dummy each
  - ▶  $D_{d,t} = 1$  if day  $t$  is weekday  $d$  (Monday, ..., Friday), else 0;  $\beta_d$ : the mean return on weekday  $d$ ;  $u_t$ : the residual
  - ▶ Newey–West HAC errors [Newey & West \(1987\)](#),  $L = 5$  days or 3 months of lags: residuals are heteroskedastic and autocorrelated

## Day-of-the-Week Effect: S&P 500 and BET



- Mean daily log return by weekday, in basis points (bp; 1 bp = 0.01%), 2000–2026, HAC 95% intervals
- S&P 500: from 0.0 bp (Friday) to 5.3 bp (Tuesday); equal-means  $p = 0.83$ : no Monday effect since 2000, unlike French (1980)
- BET: Thursday 11.2 bp ( $t = 3.09$ ), yet the joint test gives  $p = 0.31$
- Two different nulls: a day's mean equal to zero (mixes the day effect with the positive average return) vs all five means equal (the day effect itself)

## January and Turn-of-the-Month Effects

| Market         | January extra (%/month) | $t_{HAC}$ | TOM extra (bp/day) | $t_{HAC}$ | $p$    |
|----------------|-------------------------|-----------|--------------------|-----------|--------|
| S&P 500 (US)   | -0.66                   | -0.80     | 3.2                | 0.93      | 0.3539 |
| BET (Romania)  | 3.48                    | 1.49      | 17.3               | 3.62      | 0.0003 |
| Euro Stoxx 50  | -0.30                   | -0.29     | 5.1                | 1.21      | 0.2266 |
| WIG20 (Poland) | 0.53                    | 0.40      | 8.2                | 1.73      | 0.0841 |
| BUX (Hungary)  | 2.66                    | 2.25      | 15.1               | 3.63      | 0.0003 |

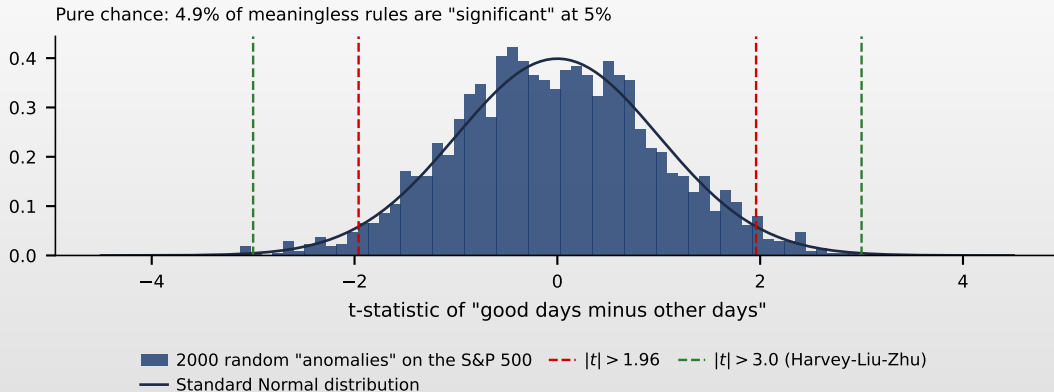
- TOM: BET +17.3 bp per day and BUX +15.1 bp, both with  $|t| > 3.5$ ; S&P 500 only +3.2 bp
- January: large but imprecise for BET ( $t = 1.49$ ); BUX, with  $t = 2.25$ , is the only market with a January-effect  $p$ -value below 0.05
- We ran 15 tests (5 markets  $\times$  3 effects): which survive a correction for multiple testing?

## Multiple Testing: Bonferroni and Holm

- **FWER** (family-wise error rate): the probability of at least one false rejection; with  $M$  independent tests at level  $\alpha$ ,  $\text{FWER} = 1 - (1 - \alpha)^M$ 
  - ▶  $M = 15$ ,  $\alpha = 5\%$ :  $\text{FWER} = 53.7\%$ , a coin flip
- **Bonferroni**: reject only if  $p_i < \alpha/M$ ; guarantees  $\text{FWER} \leq \alpha$  for any dependence
- **Holm**: sort  $p_{(1)} \leq \dots \leq p_{(M)}$ ; reject while  $p_{(j)} < \alpha/(M - j + 1)$ , stop at the first failure
  - ▶ Same guarantee, never less powerful than Bonferroni
- Our 15 calendar tests: Bonferroni threshold 0.0033; 3 raw rejections, 2 survive Holm (BET and BUX turn of the month); Holm stops at BUX January
- Seminar B6 carries out the ranking step by step; bootstrap methods that use the dependence between tests:

▶ Appendix

## How Many “Anomalies” Does Pure Chance Produce? (1/2)

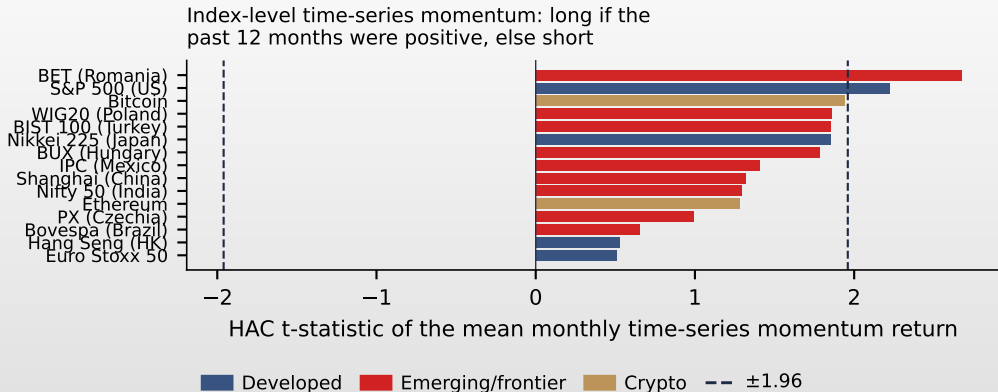


## How Many “Anomalies” Does Pure Chance Produce? (2/2)

- 2000 meaningless rules on the S&P 500: each labels a random 20% of days as “good” and tests good minus other days
- 4.9% have  $|t| > 1.96$ , the 5% the test promises; the largest  $|t|$  is 3.19
- Whoever tries many rules and reports the best is very likely to find a “significant anomaly”; data snooping in trading rules: [Sullivan, Timmermann & White \(1999\)](#)
- [Harvey, Liu & Zhu \(2016\)](#) suggest  $|t| > 3.0$  for a new factor; here 0.25% of random rules pass. A higher bar is a heuristic, not a substitute for defining the family of tests tried

 [MFM\\_ch2\\_multiple\\_testing](#)

## Momentum and Index-Level Time-Series Momentum (1/2)



## Momentum and Index-Level Time-Series Momentum (2/2)

- Reversal of 3–5-year losers De Bondt & Thaler (1985); cross-sectional momentum over 3–12 months Jegadeesh & Titman (1993) (Chapter 3)
- TSMOM Moskowitz, Ooi & Pedersen (2012): signal  $s_t = \text{sign}(\sum_{j=1}^{12} r_{t-j})$ 
  - ▶ +1 (long) if the past 12 monthly log returns sum to a gain, −1 (short) otherwise
  - ▶ strategy return  $s_t R_t$ ,  $R_t$  the index return of month  $t$ ; bars: HAC  $t$  (6 lags)
- Significant: BET ( $t = 2.68$ , 1.40% a month), S&P 500 ( $t = 2.22$ ); Bitcoin  $t = 1.94$
- Sharpe ratio below buy-and-hold: BET 0.65 vs 0.75, S&P 500 0.48 vs 0.52: much of TSMOM is the equity premium earned while long
- Data limits: price indices without financing rates or dividends, 15 markets tested; full table: [▶ Appendix](#)

## Psychology in Prices, from the Floor to the Feed



New York Stock Exchange trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain;  
Wikimedia Commons

- **Excess volatility** Shiller (1981): prices move far more than the dividends that should justify them; **overreaction** De Bondt & Thaler (1985): extreme losers become winners over 3–5 years
- **Prospect theory** Kahneman & Tversky (1979): losses loom larger than gains relative to a reference point
  - ▶ Disposition effect: winners sold too early, losers held too long
- **Overconfidence**: too much trading; **anchoring** and **limited attention**: slow reaction to news, a candidate cause of momentum
- **Herding** Pele, Mazurencu-Marinescu & Nijkamp (2013): once voices on a floor, now order books, algorithms and social media (Chapter 10)
- Biases alone do not break efficiency: rational arbitrageurs could trade against them. Can they?

## Limits to Arbitrage



Andrei Shleifer, Harvard

Photo: Manuel Castells (2018); CC BY-SA 2.0;

Wikimedia Commons

- **Shleifer & Vishny (1997)**: real arbitrage is risky and needs capital
- **Noise-trader risk**: mispricing can get worse before it corrects
  - ▶ Noise traders: investors trading on non-information (sentiment, rumours)
- **Funding constraints**: managers lose clients after losses and must close positions at the worst time
- **Costs**: short-selling limits, margins, trading costs, no perfect substitutes
- **Anomalies can persist where arbitrage is hard**: in small, illiquid markets such as BET we indeed find the strongest persistence
- **The AMH reconciles both views**: rational and biased behaviour coexist and evolve

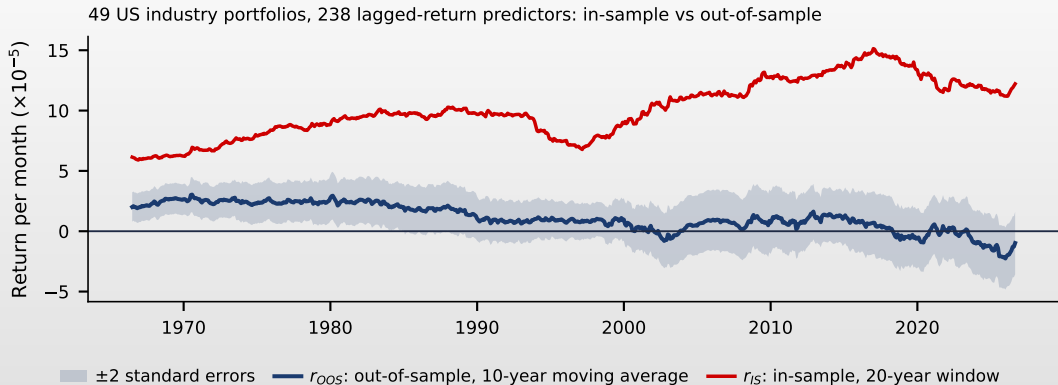
## Can Machine Learning Beat Efficient Markets?

- ▣ ML (machine learning) learns a prediction rule from data and captures nonlinear patterns that autocorrelation tests miss ([Chapter 13](#))
- ▣ [Gu, Kelly & Xiu \(2020\)](#): trees and neural networks give large economic gains over linear regressions in the US cross-section, yet their monthly out-of-sample  $R^2$  is below 1%
- ▣ Large language models read news and filings: [Tak & Pele \(2025\)](#) forecast stock trends with LLaMA
- ▣ What ML adds to the cautions of this chapter: **chronological validation**
  - ▶ train only on data available before each forecast (walk-forward); random cross-validation leaks the future (look-ahead bias)
- ▣ Key question: when many predictors are available, what does an in-sample rejection of efficiency mean? The case study answers it

## Case Study: Rational Learning with Many Predictors

- **Martin & Nagel (2022)**: investors price  $N$  assets whose cash flows depend on  $J$  characteristics with **unknown** coefficients and learn them optimally (Bayes) from past data
  - ▶ the posterior mean is a ridge regression: coefficients shrunk towards 0; investors are risk-neutral, so no predictability can be a risk premium
- **In sample**: with  $J/N \rightarrow \psi > 0$  the test of no predictability rejects with probability tending to 1 (Martin & Nagel (2022), Prop. 4)
  - ▶  $\psi$ : the limit of the number of predictors per asset,  $J/N$
  - ▶ a rejection is not evidence of irrationality: the rational-expectations null is the wrong benchmark when parameters must be learned
- **Out of sample**: a portfolio built with coefficients from other periods has  $E[r_{\text{OOS}}] = 0$  (Martin & Nagel (2022), Prop. 6)
  - ▶  $r_{\text{OOS}}$ : the return of that portfolio in a period not used to estimate its coefficients
  - ▶ out-of-sample tests keep their economic meaning: they are the right test of efficiency
- US stocks: in-sample  $R^2$  about 1%, stable; out of sample much lower and within two standard errors of 0 after 2014 (Martin & Nagel (2022), Fig. 8). Model, simulations and design: [▶ Appendix](#)

## Replication on Course Data: In Sample vs Out of Sample (1/2)

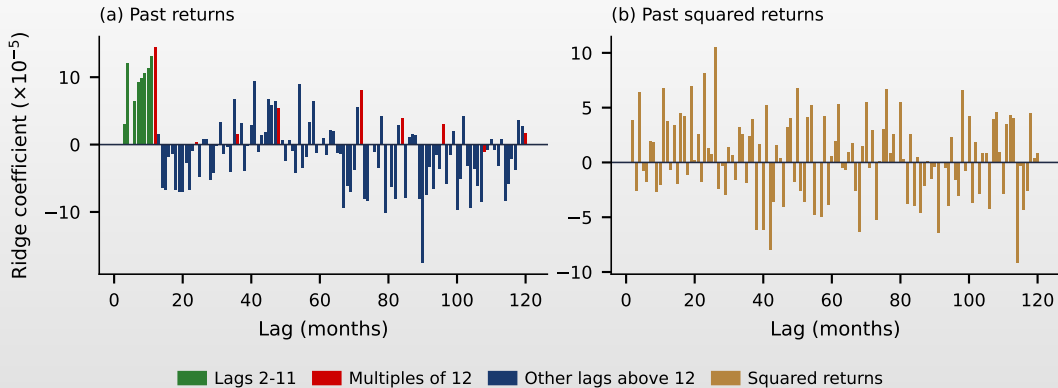


## Replication on Course Data: In Sample vs Out of Sample (2/2)

- 49 industry portfolios ([Kenneth French Data Library](#)), July 1926 – August 2026, 238 lagged-return predictors, rolling 20-year OLS, specification of [Martin & Nagel \(2022\)](#), Sec. 7
- $r_{IS}$  (in-sample return of the predicted-return portfolio): 5.90 to 15.14 ( $\times 10^{-5}$  a month), positive by construction; mean in-sample  $R^2 = 5.4\%$
- $r_{OOS}$  (out-of-sample return of the same portfolio), 842 months: mean  $1.05 \times 10^{-5}$ , about 11% of  $r_{IS}$ ; to June 1986: 4.16 SE above 0, afterwards 0.62
- Bands ( $\pm 2$  SE) are descriptive, not a test of  $E[r_{OOS}] = 0$ ; 49 portfolios instead of thousands of stocks make  $J/N$  large

 MFM\_ch2\_big\_data

## Replication on Course Data: Ridge Coefficients by Lag (1/2)



## Replication on Course Data: Ridge Coefficients by Lag (2/2)

- Full-sample ridge, January 1971 – August 2026, penalty by leave-one-year-out cross-validation (56 years); compare [Martin & Nagel \(2022\)](#), Fig. 7
- Lags 2–11: 8 of 10 positive, **momentum** [Jegadeesh & Titman \(1993\)](#); multiples of 12: 9 of 10 positive, **seasonality** [Heston & Sadka \(2008\)](#)
- Other lags above 12: 67% negative, long-run **reversal** [De Bondt & Thaler \(1985\)](#); squared returns: no dominant sign
- Cross-validated  $R^2 = 0.013\%$  (OLS:  $-2.26\%$ ): shrinkage recovers the known anomalies, yet they predict almost nothing in the years left out

 MFM\_ch2\_big\_data

## Hypothesis: Did Spot ETFs Make Bitcoin More Efficient?

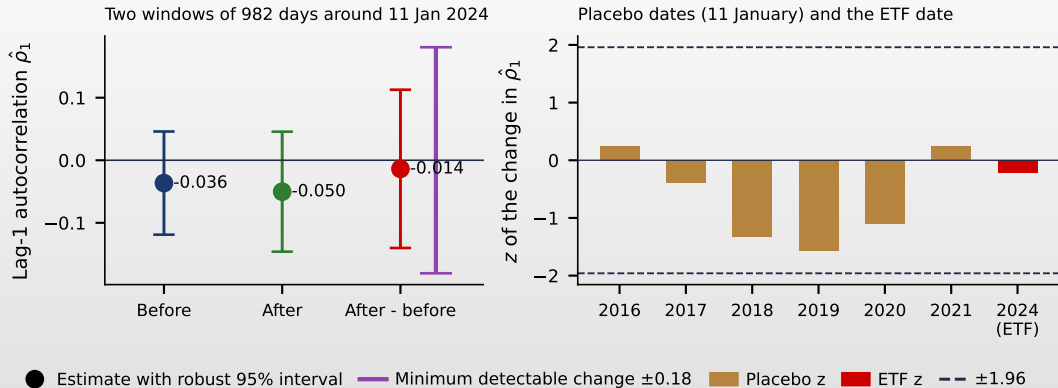
- A typical assistant proposal: “US spot Bitcoin ETFs (trading since 11 January 2024) brought arbitrageurs, so Bitcoin became more efficient”
- Discovery loop in one pass (reusable prompts: [▶ Appendix](#))
  - ▶ **literature:** [Kia, Liu, Li, Song & Xu \(2025\)](#) find that after the spot ETFs the spot market became the dominant venue for price discovery again; [Tran & Leirvik \(2020\)](#): crypto efficiency varies and has increased; resolve every DOI yourself
  - ▶ **hypothesis:**  $H_0 : \rho_{1,\text{post}} = \rho_{1,\text{pre}}$ , two windows of equal length around the regulatory date
  - ▶ **data and code:** daily Bitcoin closing price (USD); first replicate the chapter's full-sample  $\hat{\rho}_1 = -0.022$
- Checks fixed before seeing results
  - ▶ robust standard errors (not  $1/\sqrt{T}$ ); equal windows; the regulatory date only, plus placebo dates
  - ▶ a control asset for confounders (the April 2024 halving, rates, inflows)
  - ▶ a correction for every statistic tried; the minimum detectable effect

## Estimation and Power

- Test: two windows of 982 days around 11 January 2024;  $z = \Delta\hat{\rho}_1 / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$ , robust standard errors  $s$ 
  - ▶  $\Delta\hat{\rho}_1 = \hat{\rho}_{1,\text{post}} - \hat{\rho}_{1,\text{pre}}$ ;  $s_{\text{pre}}$ ,  $s_{\text{post}}$ : the robust SEs of the two estimates, independent windows
  - ▶ before (May 2021 – Jan 2024):  $\hat{\rho}_1 = -0.036$  ( $s = 0.042$ ),  $\text{VR}(5) = 1.000$  ( $z^* = 0.00$ )
  - ▶ after (Jan 2024 – Sep 2026):  $\hat{\rho}_1 = -0.050$  ( $s = 0.049$ ),  $\text{VR}(5) = 0.939$  ( $z^* = -0.64$ )
  - ▶  $\Delta\hat{\rho}_1 = -0.014$ ,  $z = -0.21$ : an unchanged  $\rho_1$  is not rejected
- **Minimum detectable effect (MDE)**: the smallest true change found with 80% power at the 5% level
  - ▶  $\text{MDE} = (z_{0.975} + z_{0.80}) \times \text{SE}(\Delta\hat{\rho}_1) \approx 2.8 \times \text{SE} = 0.18$ ,  $z_p$ : the  $p$ -quantile of the standard Normal distribution (1.96 and 0.84)
  - ▶ if Bitcoin was already close to efficient before 2024, this design could not detect a smaller improvement
- Placebo at 11 January 2016–2021: 0 of 6 significant, largest  $|z| = 1.57$
- Causality would also need a control asset (Ethereum before its own ETFs in July 2024), parallel pre-trends and no spillovers

 ai\_discovery\_case.py

## Mini-Case: Estimate, Placebo and MDE (1/2)



ai\_discovery\_case.py

## Mini-Case: Estimate, Placebo and MDE (2/2)

- Left:  $\hat{\rho}_1$  before and after, with robust 95% intervals; the difference with  $\pm$ MDE
- Right: the ETF  $z$  lies inside the range produced by six placebo dates
- “Not significant” is not “no effect”: the MDE is several times the estimate
- A better outcome: intraday price discovery across venues, not daily  $\rho_1$

 ai\_discovery\_case.py

## Your Turn: A Project Seed

- ▣ **Question:** did the spot ETFs change the time-varying efficiency of Bitcoin, relative to a control asset?
  - ▶ rolling VR and DFA (this chapter) for Bitcoin and Ethereum; Ethereum is untreated until July 2024
  - ▶ if parallel trends, no anticipation and no spillovers are not credible, report descriptive, not causal, differences
- ▣ **Replicate first:** the chapter's yearly Bitcoin autocorrelations and the mini-case numbers
- ▣ **Pre-register:**  $H_0$ ,  $H_1$ , dates, windows and corrections, committed before the results
- ▣ **Report:** robust inference, placebo tests and the MDE; the code as a Quantlet
- ▣ **Log:** prompts in AI\_USE.md; every caught error (an i.i.d. standard error, an invented reference) in AI\_ERRORS.md

## Synthesis, Check Yourself, and Next: Chapter 3

### Five steps for every efficiency claim

- ▣ **Information set:** which  $\mathcal{I}_t$  (past prices, public news)?
- ▣ **Null:** RW1, RW3 or the martingale difference, and which risk model?
- ▣ **Statistic:** its variance valid under volatility clustering
- ▣ **Uncertainty:** horizons, markets and windows tried; out-of-sample evidence
- ▣ **Economics:** rejection  $\neq$  exploitable profit  $\neq$  irrational market

### Check yourself

- ▣ Why can the S&P 500 reject RW3 and still be efficient in the economic sense?
- ▣ Why does BET's  $H(20) > 0.5$  not prove long memory?
- ▣ Compare: which of BET's persistence and Bitcoin's non-rejection would survive a change of test, window and inference?
- ▣ Next, [Chapter 3](#): risk-adjusted returns with factor models, and the factor zoo

## Appendix: Pioneers, Bachelier and Samuelson



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; [Wikimedia Commons](#)



Paul A. Samuelson

Photo: Bernard Gotfryd (1970–1975); public domain; [Wikimedia Commons](#)

- **Bachelier (1900)**: prices of French government bonds move like a random walk
  - ▶ “The mathematical expectation of the speculator is zero”
- **Samuelson (1965)**: if prices are properly anticipated, their changes are unpredictable
  - ▶ Unpredictability is a sign of a well-functioning market, not of chaos
- Both results are the ancestors of the Efficient Market Hypothesis

## Appendix: Eugene Fama



Eugene F. Fama, Nobel Prize 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0; Wikimedia

[Commons](#)

- [Fama \(1970\)](#) gave the definition used in this chapter and the three forms of efficiency
- Nobel Prize 2013, shared with Robert Shiller (a sceptic of efficiency) and Lars Peter Hansen
- Update and survey: [Fama \(1991\)](#); accessible critique: [Malkiel \(2003\)](#)

## Appendix: Data Inventory and Cleaning Sensitivity

| Group                    | Series                                                          | Period         |
|--------------------------|-----------------------------------------------------------------|----------------|
| Developed equity         | S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng                   | 2000–2026      |
| Emerging/frontier equity | BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai | 2000–2026      |
| Crypto-assets            | Bitcoin, Ethereum                                               | 2014/2016–2026 |
| Exchange rate            | EUR/RON, BNR reference rate                                     | 2005–2026      |

- Daily closing values up to 18 September 2026; sample sizes  $T$  by market: summary table on the next slide
- Removing closes equal to the previous close also drops the rare genuine zero-return day (S&P 500: 3 of about 6 700 days); this moves the test statistics by less than 0.05
- Univariate tests use each series on its own calendar; multi-series analyses align prices on common trading days before computing returns

## Appendix: Summary Table, Weak-Form Tests 2000–2026

| Market                   | $T$   | $\hat{\rho}_1$ | robust SE | $\widehat{VR}(5)$ | $z^*(5)$ | CD $p$ | Runs $z$ |
|--------------------------|-------|----------------|-----------|-------------------|----------|--------|----------|
| S&P 500 (US)             | 6 714 | −0.099         | 0.027     | 0.830             | −2.77    | 0.001  | 4.34     |
| Euro Stoxx 50            | 6 836 | −0.029         | 0.019     | 0.910             | −2.08    | 0.097  | 3.41     |
| Nikkei 225 (Japan)       | 6 545 | −0.038         | 0.024     | 0.923             | −1.48    | 0.317  | 1.97     |
| Hang Seng (HK)           | 6 583 | −0.004         | 0.024     | 0.967             | −0.64    | 0.872  | −0.17    |
| BET (Romania)            | 6 670 | 0.109          | 0.028     | 1.173             | 2.98     | 0.001  | −6.05    |
| WIG20 (Poland)           | 6 685 | 0.018          | 0.018     | 1.009             | 0.24     | 0.777  | 2.25     |
| BUX (Hungary)            | 6 666 | 0.037          | 0.024     | 1.013             | 0.26     | 0.427  | −0.42    |
| PX (Czechia)             | 6 682 | 0.052          | 0.032     | 1.041             | 0.59     | 0.350  | −2.53    |
| BIST 100 (Turkey)        | 6 692 | 0.005          | 0.024     | 1.033             | 0.64     | 0.949  | −0.63    |
| Bovespa (Brazil)         | 6 622 | −0.028         | 0.025     | 0.928             | −1.29    | 0.521  | 0.88     |
| IPC (Mexico)             | 6 689 | 0.070          | 0.019     | 1.039             | 0.93     | 0.001  | −3.27    |
| Nifty 50 (India)         | 6 606 | 0.048          | 0.024     | 1.048             | 0.91     | 0.178  | −5.46    |
| Shanghai (China)         | 6 474 | 0.021          | 0.019     | 1.046             | 1.11     | 0.409  | 1.02     |
| Bitcoin                  | 4 384 | −0.022         | 0.026     | 0.991             | −0.19    | 0.835  | 2.68     |
| Ethereum                 | 3 913 | −0.012         | 0.027     | 1.067             | 1.17     | 0.027  | 3.91     |
| EUR/RON (reference rate) | 5 352 | 0.170          | 0.036     | 1.157             | 2.19     | 0.000  | −1.66    |

□  $T$ : number of daily returns; CD: Chow–Denning over  $q = 2, 5, 10, 20$ ; EUR/RON from July 2005, crypto on 365 days per year

## Appendix: Runs Test on All Markets, Other VR Tests

- ▣ Runs  $z$  (table on the previous slide): Nifty 50  $-5.46$ , IPC  $-3.27$  (persistence in emerging markets); Ethereum  $3.91$ , significant although  $\hat{\rho}_1$  is not
- ▣ Choi (1999): an automatic VR test that chooses the horizon from the data (spectral estimate at frequency zero)
- ▣ Wild-bootstrap versions for small samples: Kim (2006); applications to Asian markets: Kim & Shamsuddin (2008)

## Appendix: Automatic Portmanteau Test of the MDH

- Escanciano & Lobato (2009):  $AQ = T \sum_{j \leq \tilde{p}} \hat{\gamma}_j^2 / \hat{\kappa}_j \rightarrow \chi_1^2$ , with  $\hat{\gamma}_j = \frac{1}{T-j} \sum_t e_t e_{t-j}$  and  $\hat{\kappa}_j = \frac{1}{T-j} \sum_t e_t^2 e_{t-j}^2$  (robust to conditional heteroskedasticity)
  - ▶  $\tilde{p} \leq 10$  maximises  $Q_p = T \sum_{j \leq p} \hat{\gamma}_j^2 / \hat{\kappa}_j$  minus a penalty that switches from BIC ( $p \ln T$ ) to AIC ( $2p$ ) when some  $\sqrt{T}|\hat{\rho}_j|$  is large
- $e_t = r_t - \bar{r}$ ;  $\hat{\gamma}_j$ : autocovariance at lag  $j$ ;  $\hat{\kappa}_j$ : its robust variance; a large  $AQ$  rejects the MDH
- $AQ$ : S&P 500 13.1 ( $p = 0.0003$ ), BET 14.9 ( $p = 0.0001$ ), Bitcoin 0.8 ( $p = 0.385$ ); 5 of 16 series reject at 5%

## Appendix: Predictive Regression, Data and Full Output

- Monthly log excess return (dividends included, minus the 3-month T-bill rate) on lagged  $dp_t$ ,  $T = 1111$  months
  - ▶ Data: [Shiller \(2026\)](#) (S&P Composite, to 2026-06), extended with our S&P 500 closes and the dividends of the SPY fund; 3-month US Treasury bill yield (FRED, series TB3MS; annual %), monthly log rate  $\ln(1 + \text{TB3MS}/1200)$  lagged one month
- OLS:  $\hat{\beta} = 0.267$ , Newey–West SE 0.293 (12 lags),  $t = 0.91$ , OLS  $t = 1.16$ ,  $R^2 = 0.12\%$
- Corrected slope / unadjusted HAC SE =  $-0.22$ : only illustrative, it ignores the uncertainty of the correction
- Economic value [Campbell & Thompson \(2008\)](#): the expected return of a mean–variance investor who times the market with the forecast rises by the proportion
  - ▶  $\frac{R^2}{1 - R^2} \cdot \frac{1 + S^2}{S^2}$ ,  $S$  the monthly Sharpe ratio; with  $S = 0.160$  and  $R^2 = 0.5\%$ : 20%
- Diebold–Mariano  $t = -0.57$  for the same forecasts, against Clark–West  $t = 0.42$

## Appendix: Harold Hurst and the Nile



Harold Edwin Hurst (1880–1978)

Photo: Elliott & Fry (1953); public domain; Wikimedia

[Commons](#)

- Hurst (1951) designed Nile reservoirs: wet years followed wet years more often than chance allows, measured with the rescaled range



Nilometer on Elephantine Island, Aswan

Photo: Olaf Tausch (2011); CC BY 3.0; Wikimedia Commons

◀ Back



## Appendix: The Anis–Lloyd Expectation

- ▣ Even for i.i.d. returns, the slope of  $\ln(R/S)$  on  $\ln n$  exceeds 0.5 for small  $n$
- ▣ Anis & Lloyd (1976): exact  $\mathbb{E}[R_n/s_n]$  under i.i.d. Gaussian returns
  - ▶  $\mathbb{E}[R_n/s_n] = \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma(n/2)} \sum_{i=1}^{n-1} \sqrt{\frac{n-i}{i}}$  ( $\Gamma$ : the gamma function)
- ▣ Corrected exponent:  $H_{AL} = 0.5 + \text{slope of } \ln(R/S) - \ln \mathbb{E}[R_n/s_n] \text{ on } \ln n$
- ▣ Uncorrected estimates overstate memory by about 0.04–0.05 with daily data

## Appendix: Long Memory with Inference, the Local Whittle Estimator

- $d = H - 0.5$ ; spectral density  $f(\lambda) \sim G\lambda^{-2d}$  as  $\lambda \rightarrow 0$  ( $G > 0$  a constant;  $d > 0$ : the density explodes at low frequencies, long memory); fit only the first  $m$  Fourier frequencies  $\lambda_j = 2\pi j/T$ 
  - ▶  $\hat{d} = \arg \min_d \left[ \ln \left( \frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j \right]$ ,  $I$ : the periodogram
  - ▶ Robinson (1995): for  $|d| < 1/2$ ,  $m \rightarrow \infty$ ,  $m/T \rightarrow 0$ , a smooth  $f$  near 0 and a linear process with conditionally homoskedastic martingale-difference innovations,  $\sqrt{m}(\hat{d} - d) \rightarrow N(0, 1/4)$  without Gaussianity: a confidence interval instead of a band
- $m = T^{0.65}$ , 95% CI  $\hat{d} \pm 1.96/(2\sqrt{m})$ :
  - ▶ S&P 500  $-0.013$   $[-0.069, 0.043]$ ; Bitcoin  $0.063$   $[-0.002, 0.127]$
  - ▶ BET  $0.118$   $[0.062, 0.174]$ ; Ethereum  $0.112$   $[0.045, 0.179]$
- Bandwidth trade-off: a large  $m$  reaches frequencies where short-run autocorrelation (BET  $\hat{\rho}_1 = 0.109$ ) biases  $\hat{d}$  upward
  - ▶ with  $m = T^{0.5}$  BET still gives  $0.135$  (SE  $0.056$ ): a sensitivity check supporting, not proving, that its  $d > 0$  is not only short memory
  - ▶ GARCH returns also strain the conditions above

## Appendix: EMH versus AMH

|                | EMH                                              | AMH                                                 |
|----------------|--------------------------------------------------|-----------------------------------------------------|
| Investors      | Rational, optimising                             | Bounded rationality, heuristics                     |
| Efficiency     | An equilibrium property, tested as a restriction | A degree that changes with conditions               |
| Risk–return    | Equilibrium risk premia, possibly time-varying   | Changes with the population of investors            |
| Anomalies      | Mispriced risk or data mining                    | Transient profit opportunities                      |
| Empirical tool | Tests of an equilibrium restriction              | Tests of time and state variation (rolling windows) |

## Appendix: Is Bitcoin Efficient? The Literature

- [Urquhart \(2016\)](#): Bitcoin inefficient over 2010–2016; some tests indicate efficiency in the later subperiod
  - ▶ His battery (Ljung–Box, runs, Bartels, automatic VR, BDS, R/S) includes BDS, which rejects under volatility clustering alone: not a test of RW3
- [Nadarajah & Chu \(2017\)](#): with a power transformation of returns, Bitcoin is weak-form efficient
- [Kristoufek \(2018\)](#): Bitcoin markets mostly inefficient in 2010–2017, with efficiency changing over time
- [Tran & Leirvik \(2020\)](#): time-varying efficiency across many crypto-assets, increasing over time
- [Pele et al. \(2023\)](#): crypto-assets have become a distinct asset class; their returns share the stylised facts of other markets
- Year by year in this chapter: Bitcoin 2011–2013  $\hat{\rho}_1 = -0.002, -0.071, 0.069$ ; the largest  $|z^*(5)|$  in any year is 1.71

## Appendix: Romania, Frontier then Emerging

- ▣ [FTSE Russell \(2019\)](#): Romania reclassified from Frontier to Secondary Emerging, announced in September 2019, effective 21 September 2020
- ▣ Expected channels: more foreign investors, more analysts, more liquidity  $\Rightarrow$  faster price discovery
- ▣ Earlier BVB evidence: crashes and log-periodic bubbles [Pele & Mazurencu-Marinescu \(2012\)](#); herding in an illiquid market [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
- ▣ Did efficiency improve? Seminar C1 compares five years before and after, with WIG20 and BUX as a comparison group and a block bootstrap
  - ▶ Block bootstrap: resamples blocks of consecutive days to keep the dependence [Künsch \(1989\)](#); a before/after comparison is descriptive unless confounders are controlled

## Appendix: How Fast Is “Fast”?

- At high frequency, futures prices react to scheduled macro news within microseconds; [Tak, Schlamp, Osterrieder & Pele \(2026\)](#) measure this race at Eurex and CME
- Crypto events: [Pele, Lin, Wang & Gaman \(2025\)](#) study market responses to Ethereum development milestones
- Speed of adjustment is itself a measure of efficiency, and it depends on technology ([Chapter 10](#), market microstructure)

## Appendix: Dependent Tests, Reality Check, SPA and StepM

- Bonferroni and Holm ignore the correlation between tests (same markets, same days)
  - ▶ [White \(2000\)](#): Reality Check, bootstrap of  $\max_j$  over all rules tried; [Hansen \(2005\)](#): SPA, studentised and less sensitive to poor rules
  - ▶ [Romano & Wolf \(2005\)](#): StepM (Romano–Wolf step-down procedure), step-down control of the FWER from the **joint** bootstrap distribution of all statistics
- Our 15 calendar tests: StepM with a bootstrap of blocks of 3 calendar months, common to all markets, 999 replications
  - ▶ Critical  $p$  at the first step 0.0016 (Bonferroni: 0.0033); 2 rejections, BET and BUX TOM, as with Holm
  - ▶ Why stricter than Bonferroni? At nominal 5%, the bootstrap rejection rates of the HAC tests are 6.0% (day of week), 6.4% (TOM), 6.4% (January)
- From over two million simulated strategies: thresholds  $|t| > 3.8$  (time series) and 3.4 (cross-section); ignoring them makes about 45% of rejections false [Chordia, Goyal & Saretto \(2020\)](#); lowering false discoveries raises missed ones [Harvey & Liu \(2020\)](#)

## Appendix: Time-Series Momentum, All Markets

| Market             | Mean (%/month) | $t_{HAC}$ | Sharpe TSMOM | Sharpe buy-and-hold | $\hat{\rho}_1$ monthly |
|--------------------|----------------|-----------|--------------|---------------------|------------------------|
| S&P 500 (US)       | 0.61           | 2.22      | 0.48         | 0.52                | 0.03                   |
| Euro Stoxx 50      | 0.16           | 0.51      | 0.11         | 0.16                | 0.05                   |
| Nikkei 225 (Japan) | 0.61           | 1.85      | 0.39         | 0.42                | 0.09                   |
| Hang Seng (HK)     | 0.18           | 0.53      | 0.10         | 0.19                | 0.04                   |
| BET (Romania)      | 1.40           | 2.68      | 0.65         | 0.75                | 0.19                   |
| WIG20 (Poland)     | 0.67           | 1.86      | 0.36         | 0.26                | 0.02                   |
| BUX (Hungary)      | 0.72           | 1.79      | 0.40         | 0.64                | 0.10                   |
| PX (Czechia)       | 0.38           | 0.99      | 0.24         | 0.45                | 0.10                   |
| BIST 100 (Turkey)  | 1.03           | 1.85      | 0.37         | 0.75                | -0.08                  |
| Bovespa (Brazil)   | 0.28           | 0.65      | 0.14         | 0.51                | 0.11                   |
| IPC (Mexico)       | 0.44           | 1.41      | 0.32         | 0.63                | 0.03                   |
| Nifty 50 (India)   | 0.50           | 1.29      | 0.28         | 0.64                | 0.02                   |
| Shanghai (China)   | 0.59           | 1.32      | 0.30         | 0.23                | 0.08                   |
| Bitcoin            | 4.47           | 1.94      | 0.72         | 1.08                | 0.13                   |
| Ethereum           | 6.50           | 1.28      | 0.57         | 0.90                | 0.16                   |

- 307 months for equities, fewer for crypto; Sharpe ratio  $\sqrt{12} \bar{x}/s_x$  of price returns ( $\bar{x}$ ,  $s_x$ : mean and SD of the monthly strategy returns), no risk-free rate subtracted; Moskowitz, Ooi & Pedersen (2012) use futures excess returns scaled to 40% ex-ante volatility

## Appendix: Robert Shiller and Richard Thaler



Robert J. Shiller, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- Shiller (1981): stock prices move far more than the dividends that should justify them (excess volatility)
- De Bondt & Thaler (1985): investors overreact; extreme losers become winners over the next 3–5 years



Richard H. Thaler, Nobel Prize 2017

Photo: Bengt Nyman (2017); CC BY 2.0; Wikimedia Commons



## Appendix: Case Study, Research Question

- [Martin & Nagel \(2022\)](#): investors price  $N$  assets whose cash flows depend on  $J$  characteristics with unknown coefficients; what does a rejected no-predictability test mean when  $J$  is comparable to  $N$ ?
  - ▶ the coefficients must be learned from past data: a high-dimensional prediction problem, as in machine learning
- Semi-strong efficiency: prices “fully reflect” public information [Fama \(1970\)](#); two readings
  - ▶ investors know the parameters of the cash-flow model (rational expectations: the null of most efficiency tests)
  - ▶ investors update their beliefs optimally (Bayes) with the data available in real time
- Why it matters: hundreds of in-sample return predictors, the “factor zoo” [Harvey, Liu & Zhu \(2016\)](#); text, satellite images and social media keep increasing  $J$ 
  - ▶ does every significant predictor need a risk-based or a behavioural explanation?
- Link to this chapter: in-sample vs out-of-sample predictability [Campbell & Thompson \(2008\)](#), [Goyal, Welch & Zafirov \(2024\)](#), and the decay of anomalies after publication [McLean & Pontiff \(2016\)](#)
- Investors are risk-neutral in the model: no joint-hypothesis problem, so no predictability can be a risk premium

## Appendix: Case Study, Model and Method (1/2)

### □ **Model** (Martin & Nagel (2022), Sec. 2)

- ▶ section, equation, proposition and figure numbers in this appendix are those of the paper
- ▶ cash-flow growth  $\Delta y_t = Xg + e_t$ , with  $e_t \sim N(0, I)$  and a zero interest rate
- ▶  $\Delta y_t$ : the  $N$  cash-flow growth rates;  $X$ : the  $N \times J$  matrix of characteristics;  $g$ : the  $J$  unknown coefficients;  $I$ : identity matrix

### □ **Learning** (Prop. 1–2)

- ▶ prior  $g \sim N(0, (\theta/J) I)$ , the true distribution from which  $g$  is drawn;  $\theta$ : signal strength
- ▶ the posterior mean is a ridge regression (OLS with a penalty on the squared coefficients): coefficients shrunk towards 0
- ▶ realised return = underreaction caused by shrinkage + estimation noise + unpredictable shock

### □ **Asymptotics and simulations** (Sec. 4)

- ▶  $N, J \rightarrow \infty$  with  $J/N \rightarrow \psi > 0$
- ▶ simulations with  $N = 1000$ , entries of  $X$  drawn from the standard Normal distribution and  $\theta = 1$

## Appendix: Case Study, Model and Method (2/2)

### □ Tests

- ▶ in sample: a cross-sectional OLS of  $r_{t+1}$  on  $X$  (requires  $J < N$ ) with a joint  $\chi^2$  test
- ▶ out of sample: a portfolio with weights  $\frac{1}{N}Xh_{s+1}$ , from coefficients estimated in other periods
- ▶  $r_{\text{OOS},t+1} = \frac{1}{N}h'_{s+1}X'r_{t+1}$ ,  $s \neq t$  (eq. 19);  $r_{1s}$ : the same with  $s = t$
- ▶  $h_{s+1}$ : the estimated coefficient vector;  $r_{t+1}$ : the  $N$  returns

### □ Empirical design (Sec. 7)

- ▶ US stocks in CRSP, without those below the 20th NYSE size percentile or priced below \$1
- ▶ predictors: simple and squared monthly returns at lags  $t - 2$  to  $t - 120$  (238 predictors), demeaned and standardised across stocks every month
- ▶ full-sample ridge, 1971 – June 2019, penalty chosen by leave-one-year-out cross-validation
- ▶ rolling 20-year OLS gives  $r_{\text{OOS}}$  from January 1956

## Appendix: Case Study, Formal Results

- In sample (numbering as in [Martin & Nagel \(2022\)](#)): the rejection probability of the no-predictability test tends to 1 as  $N, J \rightarrow \infty$  (Prop. 4); for  $\psi > 0.4$  ( $\theta = 1, t = 1$ ) the probability of no rejection is of the order  $e^{-0.015N}$  (Fig. 2)
  - ▶ with fully rational Bayesian investors, the rational-expectations null is almost surely rejected: a rejection is not evidence of irrationality
- Simulations ( $N = 1000$ , four periods of learning): a test of nominal size 5% rejects with probability above 90% once  $J > N/2$  (Fig. 3b)
  - ▶ not a size distortion: the in-sample predictable component really is in the prices
- Single-predictor tests with  $k$ -FWER control (probability of  $k$  or more false rejections at most 5%): for  $k = 20$  and  $J/N = 0.05$ , the probability of at least  $k$  rejections is 25% instead of 5%; about 80% for  $J/N$  close to 1 (Fig. 4)
  - ▶ multiple-testing corrections do not help: the problem is the null hypothesis, not the test
- Out of sample:  $E[r_{\text{OOS}, t+1}] = 0$ , forward and backward in time (Prop. 6); positive only if investors shrink more than the data warrant (Fig. 6)
  - ▶ out-of-sample tests keep their economic meaning
- US stocks: the ridge coefficients recover momentum, long-run reversal and momentum seasonality (Fig. 7);  $r_{15} \approx 0.006\%$  per month, an in-sample  $R^2$  of about 1%, stable over time (Fig. 8)
  - ▶  $r_{\text{OOS}}$  is much lower and often near 0; in the 10-year windows ending in 2014–2019 it is less than two standard errors above 0: little of the in-sample predictability survives

## Appendix: Case Study, Replication, Limitations and Extensions

- **Replicate with the course data** (stock-level CRSP files are not part of them)
  - ▶ 49 industry portfolios, monthly value-weighted returns from July 1926 ([Kenneth French Data Library](#)); the same 238 predictors, monthly standardisation and rolling 20-year windows (Seminar 2, C3)
  - ▶ data limitation: 49 portfolios instead of thousands of stocks; the size and price filters do not apply to portfolios
- **Limits stated by the authors**
  - ▶ lagged returns are equilibrium outcomes, not the exogenous  $X$  of the model: the application illustrates the wedge, it does not test the model
  - ▶ a  $t$ -test on the monthly  $r_{\text{OOS}}$  is not a valid test of  $E[r_{\text{OOS}}] = 0$ : the expectation also averages over  $g$
  - ▶ the backward result may rely on i.i.d. cash-flow growth; homogeneous investors and constant characteristics
- **Open extensions**
  - ▶ time series, as the authors suggest: does the dividend–price regression of this chapter show the same in-sample/out-of-sample wedge?
  - ▶ nonlinear models (trees, neural networks [Gu, Kelly & Xiu \(2020\)](#)) add functional flexibility without necessarily changing  $J$ : does the wedge grow?
  - ▶ AMH: is early out-of-sample predictability the excess shrinkage of investors with less computing power?

## Appendix: Case Study, In Sample vs Out of Sample in Detail

- $r_{IS}$  is positive in every 20-year window, as the in-sample OLS fit guarantees ( $r_{IS} = b'G^{-1}b/240 \geq 0$ ,  $G = \sum_s X_s'X_s/N_s$  and  $b = \sum_s X_s'r_s/N_s$  over the window): from 5.90 to 15.14 ( $\times 10^{-5}$  per month), an average in-sample  $R^2$  of 5.4%
  - ▶ about 1% for individual stocks in [Martin & Nagel \(2022\)](#): 238 predictors against only 40–49 portfolios a month, pooled over 240 months, inflate the in-sample fit
- Out of sample, July 1956 – August 2026 (842 months): mean  $r_{OOS} = 1.05 \times 10^{-5}$ , about 11% of the mean  $r_{IS}$
- Until June 1986: mean  $r_{OOS} = 1.98 \times 10^{-5}$ , 4.16 standard errors above 0; afterwards  $0.36 \times 10^{-5}$ , only 0.62 standard errors
  - ▶ the 10-year average is last above two standard errors in the window ending in December 1987; in the windows ending in 2014–2019 its largest ratio to the standard error is 1.25, as for stocks
- Interpretation suggested by [Martin & Nagel \(2022\)](#): the gap is consistent with investor learning, but does not establish it or rule out risk premia; early out-of-sample gains are also consistent with excess shrinkage before the computer era (Fig. 6)
- The bands are descriptive: a  $t$ -test on the monthly  $r_{OOS}$  is not a valid test of  $E[r_{OOS}] = 0$

## Appendix: The Discovery Loop, Reusable Prompts (1/2)

### □ Literature

- ▶ List peer-reviewed studies since 2020 on how ETF introductions change the efficiency or price discovery of the underlying. For each: DOI, design (event study, difference-in-differences (DiD), structural break), efficiency measure. Flag any entry you are unsure exists.
- ▶ Resolve every DOI on Crossref; the title and authors must match

### □ Hypothesis

- ▶ Give three mechanisms by which spot ETFs could change Bitcoin's serial dependence (arbitrage, investor mix, trading hours). For each, the sign of the change in  $\rho_1$  and  $VR(q)$ , and a variable that separates it from the others.

### □ Data and code

- ▶ Read the daily Bitcoin closing prices; log returns;  $\rho_1$  with a heteroskedasticity-robust standard error and the Lo-MacKinlay  $VR(5)$  with  $z^*$ , in two windows of equal length around 11 January 2024.
- ▶ Replicate a known number first: the chapter's full-sample Bitcoin  $\hat{\rho}_1 = -0.022$  (robust SE 0.026)

## Appendix: The Discovery Loop, Reusable Prompts (2/2)

### ▣ Robustness

- ▶ Write a pre-registered robustness plan: window lengths, weekday-only closes,  $VR(q)$  for  $q = 2$  to 20 with Chow-Denning, placebo dates, and Ethereum as a control until its own spot ETFs (July 2024).

### ▣ Critique

- ▶ Act as a hostile referee: if Bitcoin was already efficient before 2024, what could this test ever detect? Compute the minimum detectable change in  $\rho_{01}$  and name a better outcome variable.

### ▣ Report

- ▶ Estimates, robust standard errors, the placebo distribution, every test run (with a multiple-testing correction)
- ▶ Prompts that shaped a decision go into AI\_USE.md

- ▣ The assistant drafts the tests; the human decides which one answers the question

## Appendix: What the Researcher Must Check

- **i.i.d. standard errors:**  $1/\sqrt{T}$  rejects too often under volatility clustering
  - ▶ test: robust SE and  $z^*$ , as in the weak-form tests of this chapter
- **Unequal windows:** different power before and after; test: equal lengths, fixed in advance
- **Event date chosen from the data**
  - ▶ test: the regulatory date only, plus a placebo distribution at dates without an event
- **Confounding:** before/after is not causal; test: a control asset (Ethereum before July 2024) in a difference-in-differences
  - ▶ causal only with parallel pre-trends, no anticipation and no spillover from Bitcoin ETFs to Ethereum; the common post-period (Jan–Jul 2024) is short; otherwise the comparison is descriptive
- **Many statistics:**  $\rho_1$ , VR(2–20), runs; test: Chow–Denning or Holm, not the best single  $p$ -value
- **Low power:** report the minimum detectable change,  $\approx 2.8$  standard errors of the difference

## Appendix: Further Study on Quantinar

### Advanced courses on quantinar.com

-  Efficiency of cryptocurrency markets - A GMM-based analysis
-  Generalized Method of Moments

### Code, data and textbooks

- All charts and tables: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_02](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_02)
- Textbooks: Campbell, Lo & MacKinlay (1997), Ch. 2; Franke, Härdle & Hafner (2019)

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